

Computer Science

CHAPTER 2: Data transmission

2.1 Types and methods of data transmission

2.2 Methods of error detection

2.3 Encryption

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Learning Objectives

2.1 Types and methods of data transmission

Candidates should be able to:

- 1 (a) Understand that data is broken down into packets to be transmitted
- (b) Describe the structure of a packet
- (c) Describe the process of packet switching

Notes and guidance

- A packet of data in a unit of data contains a
 - packet header
 - payload
 - trailer
- The packet header includes the:
 - destination address
 - packet number
 - originator's address
- Data is broken down into packets
- Each packet could take a different route
- A router controls the route a packet takes
- Packets may arrive out of order
- Once the last packet has arrived, packets are reordered

Learning Objectives

2.1 Types and methods of data transmission

- 2 (a) Describe how data is transmitted from one device to another using different methods of data transmission
 - Including:
 - serial
 - parallel
 - simplex
 - half-duplex
 - full-duplex
 - Including the advantages and disadvantages of each method
- (b) Explain the suitability of each method of data transmission, for a given scenario
- 3 Understand the universal serial bus (USB) interface and explain how it is used to transmit data
 - Including the benefits and drawbacks of the interface

Learning Objectives

2.2 Methods of error detection

Candidates should be able to:

- 1 Understand the need to check for errors after data transmission and how these errors can occur
- 2 Describe the processes involved in each of the following error detection methods for detecting errors in data after transmission: parity check (odd and even), checksum and echo check
- 3 Describe how a check digit is used to detect errors in data entry and identify examples of when a check digit is used, including international standard book numbers (ISBN) and bar codes
- 4 Describe how an automatic repeat query (ARQ) can be used to establish that data is received without error

Notes and guidance

- Errors can occur during data transmission due to interference, e.g. data loss, data gain and data change
- Including parity byte and parity block check
- Including the use of:
 - positive/negative acknowledgements
 - timeout

Learning Objectives

2.3 Encryption

Candidates should be able to:

- 1 Understand the need for and purpose of encryption when transmitting data
- 2 Understand how data is encrypted using symmetric and asymmetric encryption

Notes and guidance

- Asymmetric encryption includes the use of public and private keys

Learning Objectives

2.1 Types and methods of data transmission

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2.1 Types and methods of data transmission

2.1.1 Data packets

<https://www.youtube.com/watch?v=rpOfxVgNsa0>



2.1 Types and methods of data transmission

2.1.1 Data packets

Data packets

When data is sent over long distances, it is normally broken down into packets. Packets tend to be quite small – e.g., 64KiB.

Advantages of packets

- Smaller packets are easier to control than a single long stream of data.
- Each packet can take its own route to its destination.
- This adds redundancy to data transmission if a route goes down or becomes busy.

Disadvantages of packets

- Packets require careful construction and additional information so they can be reassembled correctly at the destination.

2.1 Types and methods of data transmission

2.1.1 Data packets

Structure of a packet

Each data packet contains three main parts.



Packet header

- Destination IP address
- Sender's IP address
- Packet number – e.g., 1 of 5



Payload

- The actual data being transmitted – e.g., ~64KiB



Trailer

- Marks the end of the packet

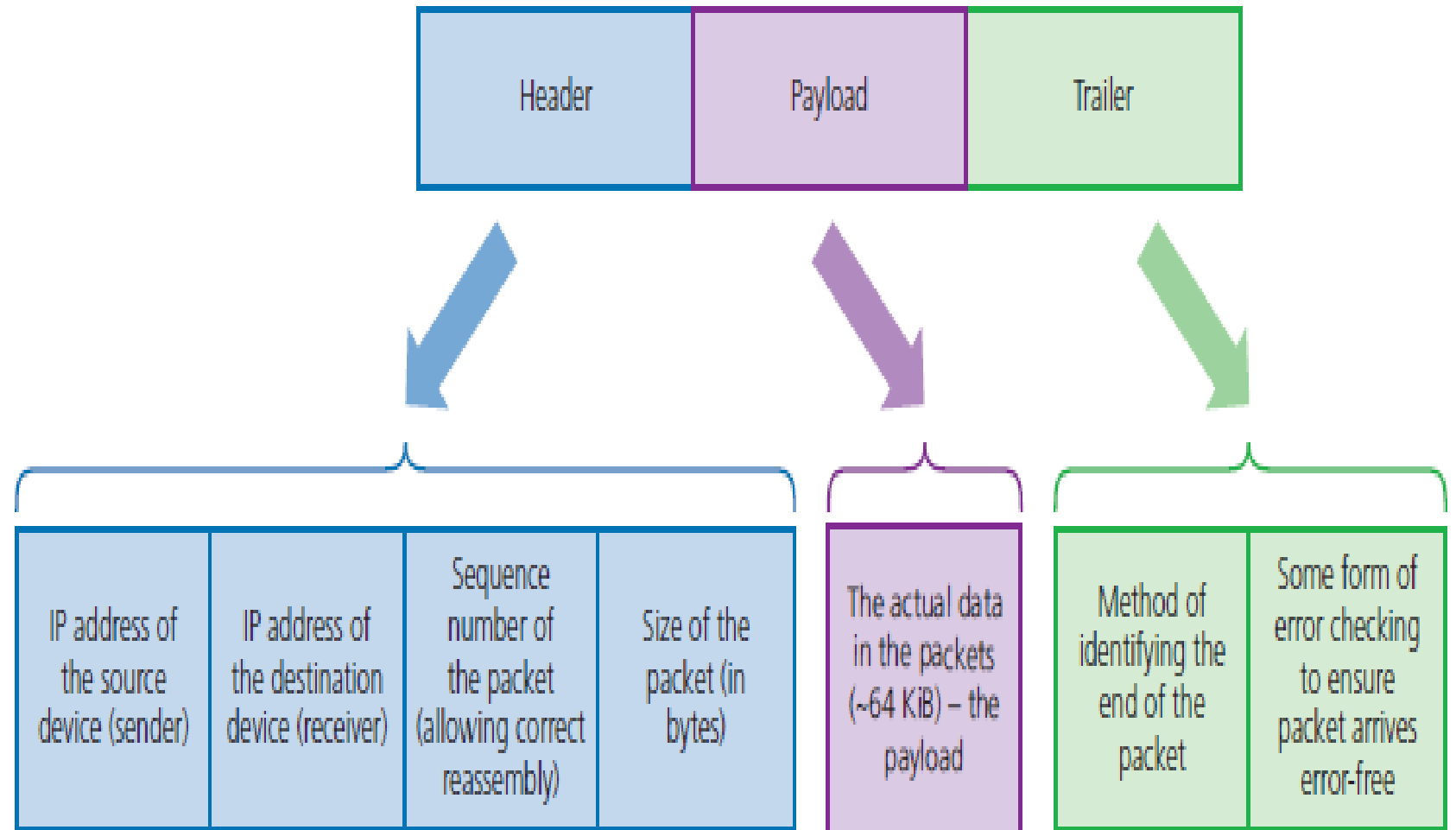
2.1 Types and methods of data transmission

2.1.1 Data packets

Packet structure

A typical packet is split up into:

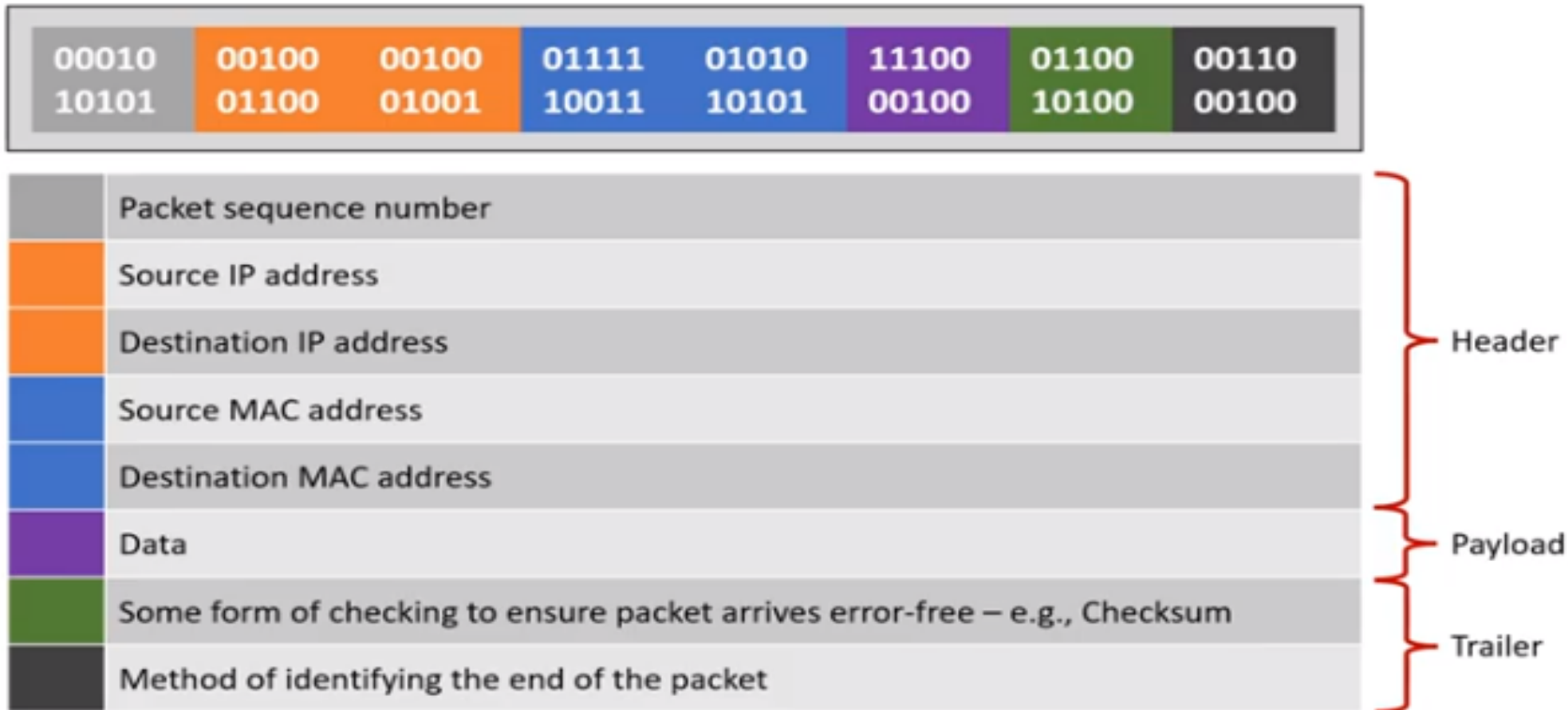
- » a packet header
- » the payload
- » a trailer.



2.1 Types and methods of data transmission

2.1.1 Data packets

The structure of packets is a lot more complicated than we have presented so far.



2.1 Types and methods of data transmission

2.1.1 Data packets



Packet and packet structure

When data is sent over long distances it is normally broken down into packets. Packets tend to be quite small.

Packets are split into three main parts:

- **Packet header:**
 - Destination IP address
 - Sender's IP address
 - Packet number – e.g., 1 of 5
- **Payload:**
 - The actual data being transmitted
- **Trailer:**
 - Marks the end of the packet

2.1 Types and methods of data transmission

2.1.2 Packet switching

<https://www.youtube.com/watch?v=PVbuTBS4Sqc>



2.1 Types and methods of data transmission

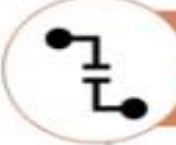
2.1.2 Packet switching



2.1 Types and methods of data transmission

2.1.2 Packet switching

Advantages of packet switching

-  No need to tie up a single communication channel
-  Easy to overcome busy or faulty lines by re-routing packets
-  Relatively easy to expand package usage
-  Possible to achieve a high data transmission rate

2.1 Types and methods of data transmission

2.1.2 Packet switching

Disadvantages of packet switching

-  Packet can become lost en route and need to be re-sent
-  Real-time streaming is more prone to errors
-  Delays at the destination while packets are re-ordered

2.1 Types and methods of data transmission

2.1.2 Packet switching



Packet switching

- Packet switching is a five-stage process:
 1. Data is broken down into packets.
 2. Each packet is assigned a sequence number.
 3. Each packet can take its own route based on congestion.
 4. Routers are responsible for controlling the route each packet takes.
 5. Packets may arrive out of order. Once the final packet arrives, they are reordered.

2.1 Types and methods of data transmission

2.1.2 Packet switching

Packet switching is a connectionless network switching technique. Here, the message is divided and grouped into a number of units called packets that are individually routed from the source to the destination. There is no need to establish a dedicated circuit for communication.

Process

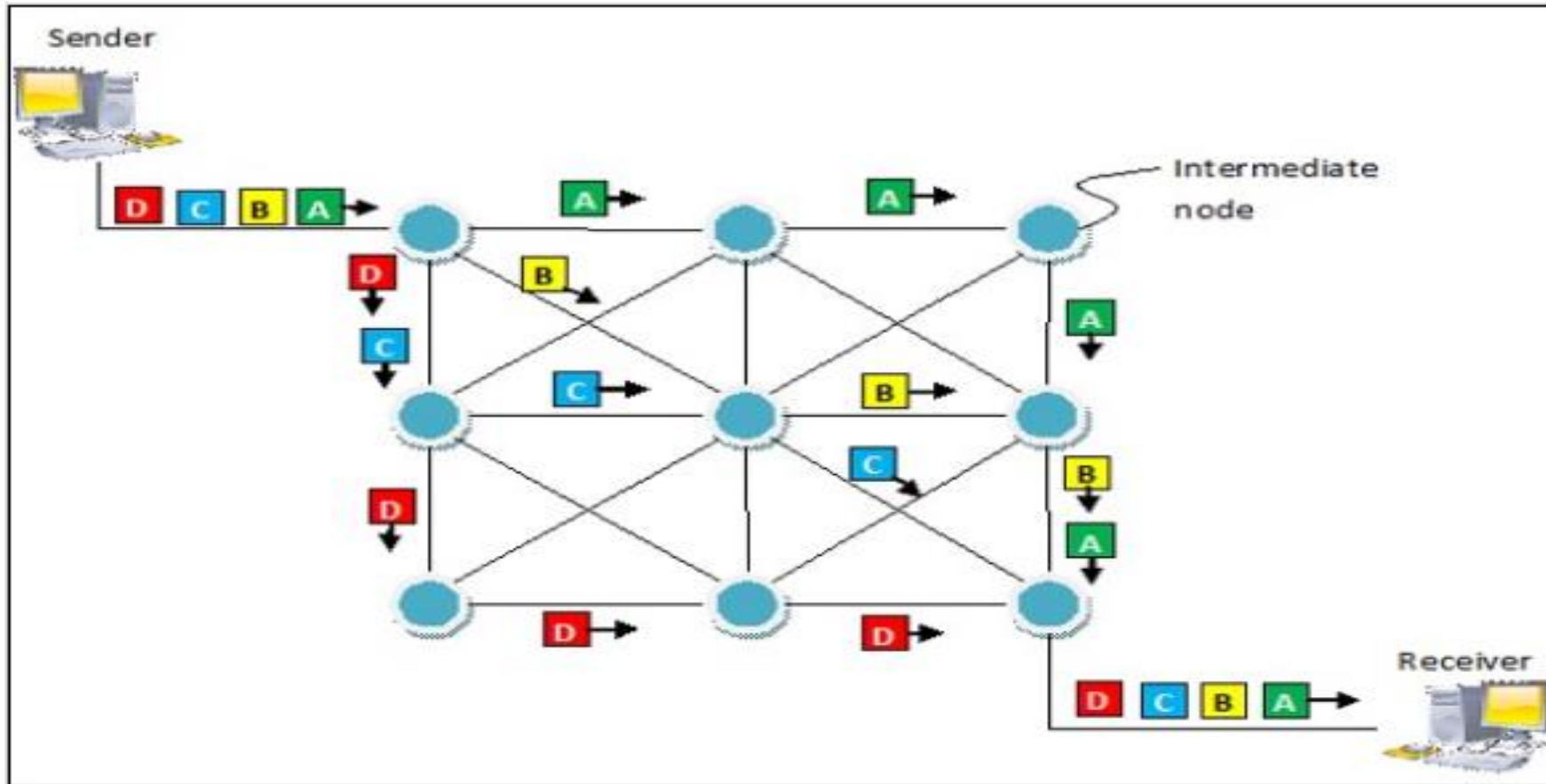
Each packet in a packet switching technique has two parts: a header and a payload. The header contains the addressing information of the packet and is used by the intermediate routers to direct it towards its destination. The payload carries the actual data.

A packet is transmitted as soon as it is available in a node, based upon its header information. The packets of a message are not routed via the same path. So, the packets in the message arrives in the destination out of order. It is the responsibility of the destination to reorder the packets in order to retrieve the original message.

The process is diagrammatically represented in the following figure. Here the message comprises of four packets, A, B, C and D, which may follow different routes from the sender to the receiver.

2.1 Types and methods of data transmission

2.1.2 Packet switching



2.1 Types and methods of data transmission

2.1.2 Packet switching

Advantages and Disadvantages of Packet Switching

Advantages

Delay in delivery of packets is less, since packets are sent as soon as they are available.

Switching devices don't require massive storage, since they don't have to store the entire messages before forwarding them to the next node.

Data delivery can continue even if some parts of the network faces link failure. Packets can be routed via other paths.

It allows simultaneous usage of the same channel by multiple users.

It ensures better bandwidth usage as a number of packets from multiple sources can be transferred via the same link.

Disadvantages

They are unsuitable for applications that cannot afford delays in communication like high quality voice calls.

Packet switching high installation costs.

They require complex protocols for delivery.

Network problems may introduce errors in packets, delay in delivery of packets or loss of packets. If not properly handled, this may lead to loss of critical information.

2.1 Types and methods of data transmission

2.1.2 Packet switching

<https://www.youtube.com/watch?v=80XDMwDthuE>

Extension video Level for Packets and packet switching



2.1 Types and methods of data transmission

2.1.2 Packet switching

Example 1:

How would packet switching be used to download a page from a website?

Solution

Answers might include

- the web page is divided up into data packets
- each packet has a header, which includes the IP address of the destination
- the router checks the header against values stored in the routing table ...
- ... to determine which router the packet needs to be sent to next (hopped) ...
- ... the MAC address of the next router is added to the package header
- the hop value is checked to see if it equals zero
- each packet may go by a different route
- the destination computer reassembles the packets building up the final web page.

2.1 Types and methods of data transmission

2.1.2 Packet switching

Example 2:

Suppose we are carrying out video conferencing using packet switching as the method of routing data. The performance of the communication is not very good.

- a) Describe some of the poor performance you might expect.
Give a reason for this poor performance.

Solution

- a) Answers might include: picture and sound may not be in synchronisation (packets arriving at different times); video not continuous – pauses (time delay in reassembling the packets of data); degraded quality of sound and video (possibly caused by competing traffic in communication lines); possible drop out (packets take different routes, so it is possible for packets to be lost).

2.1 Types and methods of data transmission

IGCSE Past Exam Paper Question 1:

2 A company has a website that is stored on a web server.

(a) The website data is broken down into packets to be transmitted to a user.

Describe the structure of a data packet.

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[4]

2.1 Types and methods of data transmission

IGCSE Past Exam Paper Answer 1:

Question	Answer	Marks
2(a)	Any four from: • it has a header • ... that contains the destination address • ... that contains the packet number • ... that contains the originator's address • it has a payload • it has a trailer.	4

2.1 Types and methods of data transmission

IGCSE Past Exam Paper Question 2:

- 6 (a) Explain how packet switching is used to transfer messages across the internet.

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2.1 Types and methods of data transmission

IGCSE Past Exam Paper Question 2:

(b) Outline the function of a router in packet switching.

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2.1 Types and methods of data transmission

IGCSE Past Exam Paper Answer 2:

Question	Answer	Marks
6(a)	One mark for each correct marking point (Max 5) • A large message is divided up into a group of smaller chunks of the same size called packets • The packet has a header and a payload • The header contains a source IP address, destination IP address (and sequence number) • Each packet is dispatched independently • ... and may travel along different routes / paths • The packets may arrive out of order • ... and are reassembled into the original message at the destination • If packets are missing / corrupted a re-transmission request is sent.	5
6(b)	One mark for each correct marking point (Max 3) • The router examines the packet's header • It reads the IP address of the destination (from the packet header) • A router has access to a routing table • ...containing information about, e.g., available hops / netmask / gateway used • ... and the status of the routes along the route • ... the router decides on the next hop / best route • ... and sends the packet on its next hop.	3

Learning Objectives

2.1 Types and methods of data transmission

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 - half-duplex
 - full-duplex
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2.1 Types and methods of data transmission

<https://www.youtube.com/watch?v=0cIWq4MZmxw>

2.1.2 Data transmission



2.1 Types and methods of data transmission

2.1.2 Data transmission

- When data is sent from one device to another and Data transmission can be either over a short distance (for example, from computer to printer) or over longer distances (for example, over a telephone network). Essentially, three factors need to be considered when transmitting data (each factor has to be agreed by both sender and receiver for this to work without error):
- the direction of the data transmission (i.e. in one direction only or in both directions)
- the method of transmission (how many bits are sent at the same time)
- the method of synchronization (activity) between the two devices.

2.1 Types and methods of data transmission

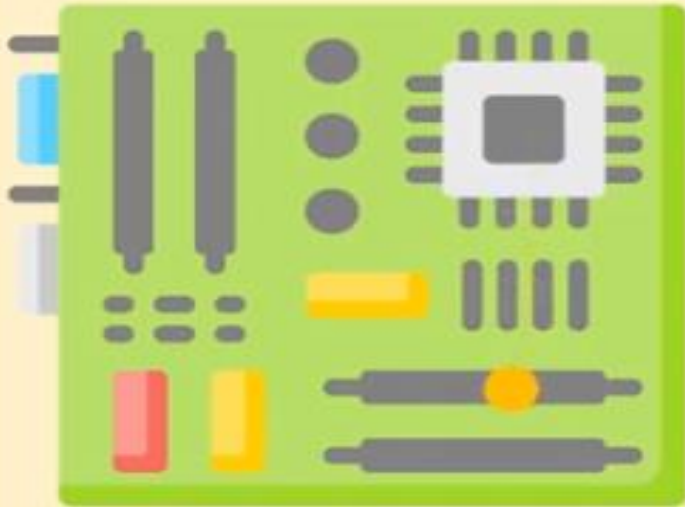
2.1.2 Data transmission

Data transmission

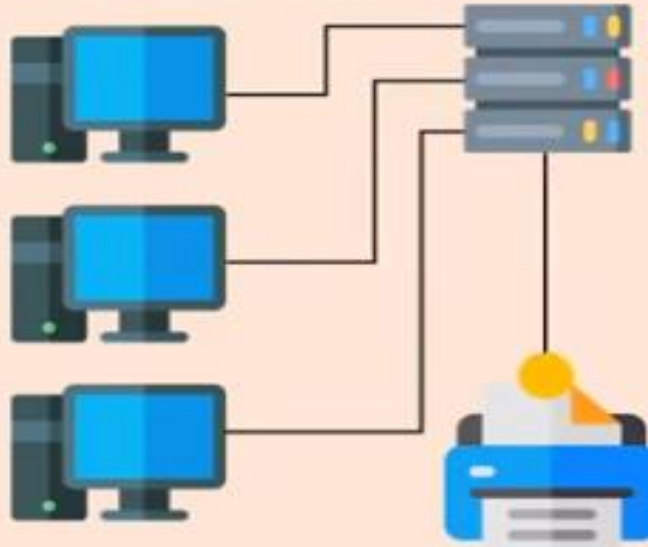
Data transmission can occur over:

- Very short distances – e.g., from one component of a computer to another.
- Short distances – e.g., from a computer to a local printer.
- Long distances – e.g., from one computer to another over a global network.

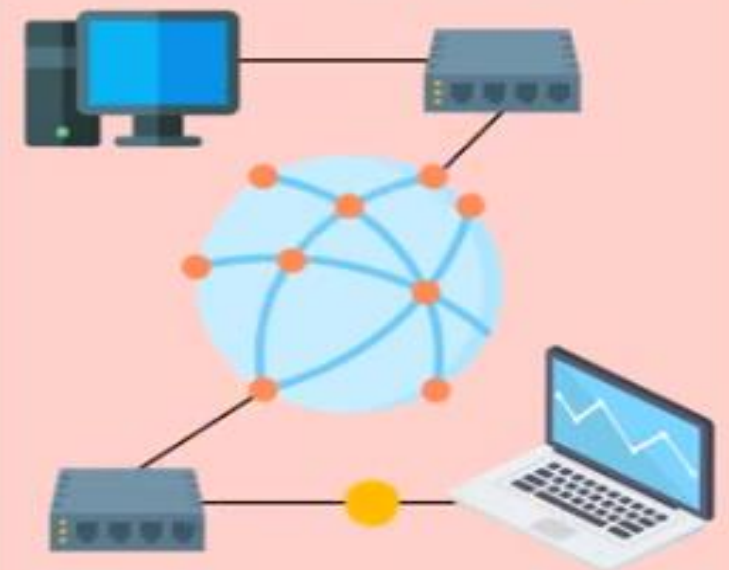
Very short-distance data transmission



Short-distance data transmission



Long-distance data transmission



2.1 Types and methods of data transmission

2.1.2 Data transmission

- SIMPLEX DATA TRANSMISSION is in one direction only (i.e. from sender to receiver). Example: data being sent from a computer to a printer.
- HALF-DUPLEX DATA TRANSMISSION is in both directions but not at the same time (i.e. data can be sent from 'A' to 'B' or from 'B' to 'A' along the same line, but not at the same time). Example: a phone conversation between two people where only one person speaks at a time.
- FULL-DUPLEX DATA TRANSMISSION is in both directions simultaneously (i.e. data can be sent from 'A' to 'B' and from 'B' to 'A' along the same line, both at the same time). Example: broadband connection on a phone line.

Watch video for Simplex, half duplex, and full duplex

https://www.youtube.com/watch?v=KDo_789WlrY



2.1 Types and methods of data transmission

2.1.2 Data transmission

Data transmission

When data is being transmitted, we need to consider three essential factors.

- Direction of data transmission – e.g., can data be transmitted in both directions or only one?

Factor 1



- Method of transmission – e.g., how many bits can be transmitted at once?

Factor 2



- How will the data be synchronised?

Factor 3

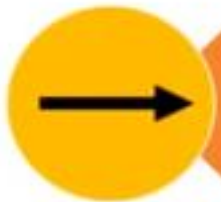
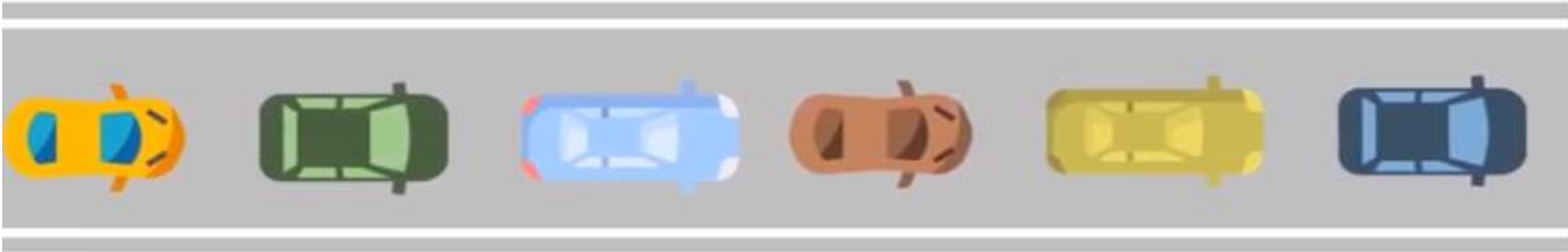


2.1 Types and methods of data transmission

2.1.2 Data transmission

Simplex transmission

Data travels in only one direction via a single cable.



This type of transmission might be used to send data from a computer to a monitor, as traffic only needs to move in one direction.

2.1.2 Data transmission

SIMPLEX DATA TRANSMISSION

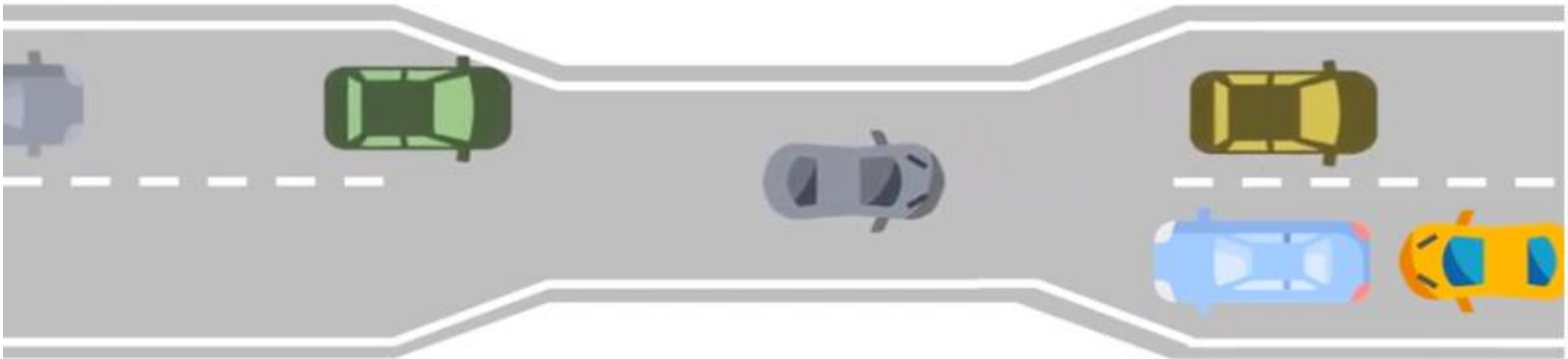
- Simplex: Data can only be sent in one direction at a time.
- For examples:
- Transfer data from a computer to a printer.
- Microphone to computer.
- Sensor to computer.
- Computer to printer/speaker/monitor.

2.1 Types and methods of data transmission

2.1.2 Data transmission

Half-duplex transmission

Data can travel in both directions via a single cable – but not at the same time.



This type of transmission occurs when using parallel printer cables. A file is sent to the printer, and the printer can respond with error codes such as "Ink low". However, the printer needs to wait until the computer has finished sending the file.

2.1.2 Data transmission

HALF-DUPLEX DATA TRANSMISSION

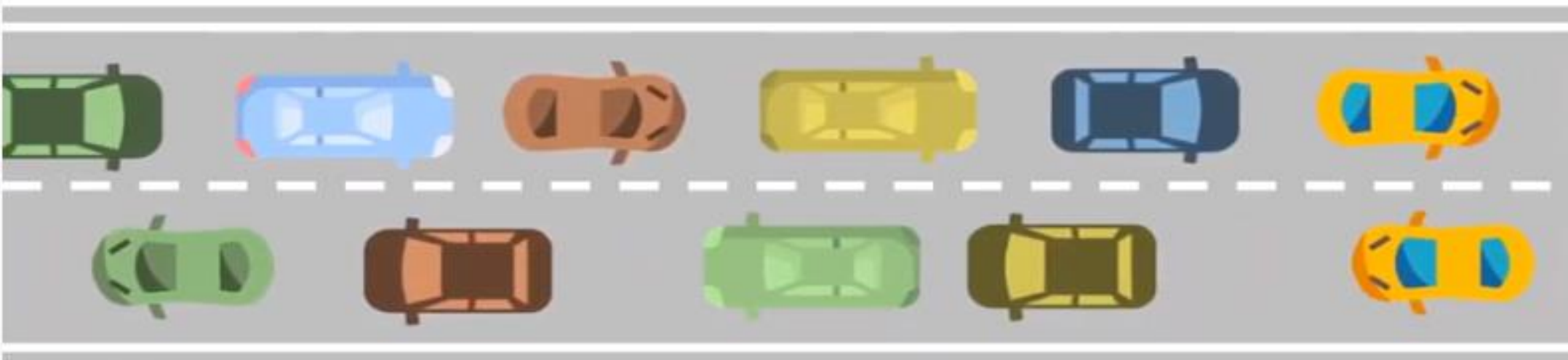
- Half duplex: Data is sent in both directions but only one direction at a time.
- Example: A two-way radio (walkie-talkie) that can receive and transmit messages, but not at the same time.

2.1 Types and methods of data transmission

2.1.2 Data transmission

Full-duplex transmission

Data can travel in both directions at the same time using two communication channels.



This is the most common form of data transmission and is typically used in network cables, both within LANS and across the internet. With these cables, data can be sent and received at the same time.

2.1.2 Data transmission

FULL-DUPLEX DATA TRANSMISSION

- Full Duplex Data Transmission: Data can be sent in both directions at the same time.
- Can be either serial or parallel.

Example:

- A telephone that can receive and transmit audio signals simultaneously.
- Instant messaging
- broadband connection
- video conferencing.

2.1 Types and methods of data transmission

2.1.2 Data transmission

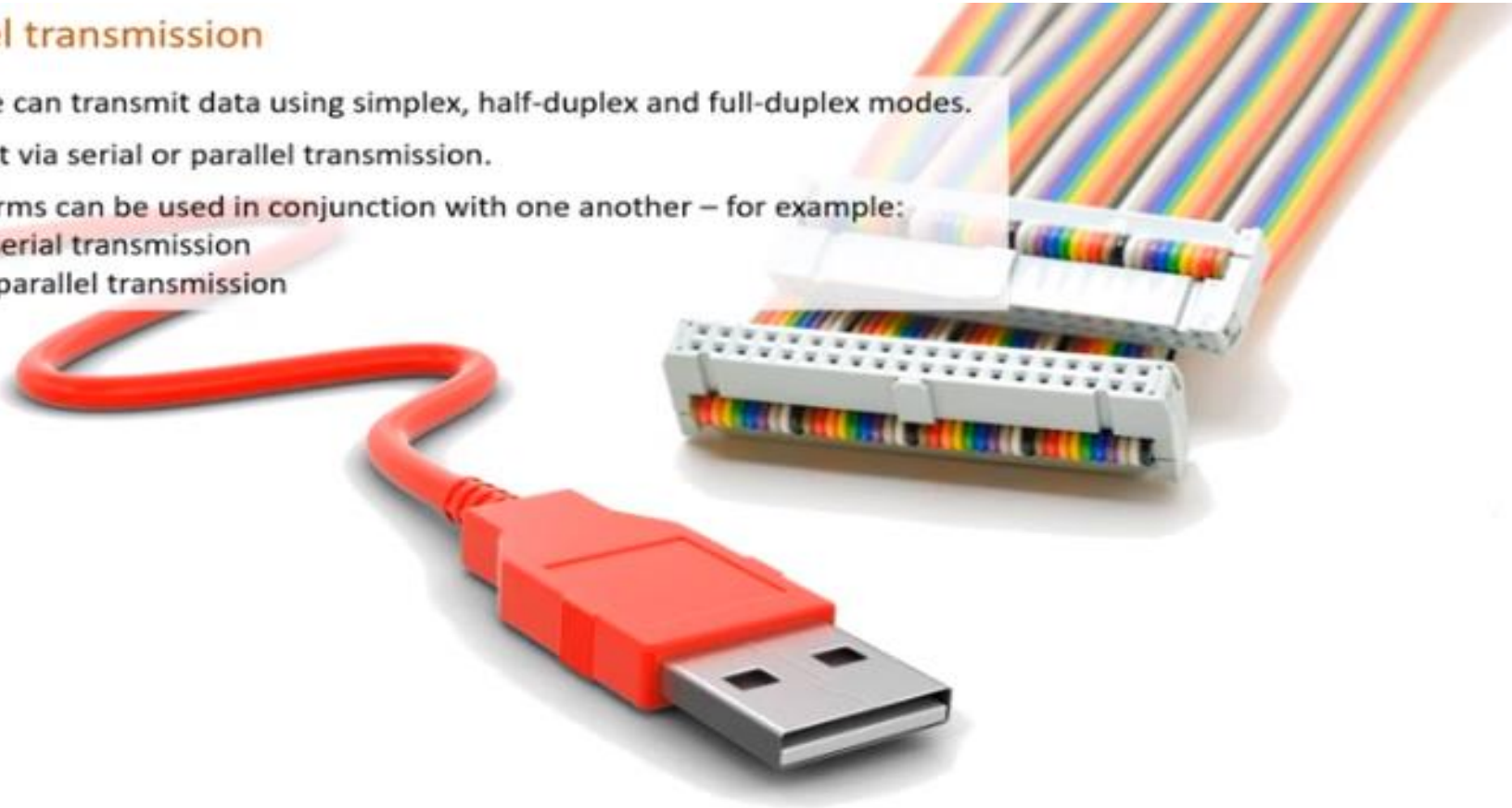
Serial vs parallel transmission

We already know we can transmit data using simplex, half-duplex and full-duplex modes.

■ Data can also be sent via serial or parallel transmission.

These two sets of terms can be used in conjunction with one another – for example:

- Full-duplex serial transmission
- Half-duplex parallel transmission



2.1.2 Data transmission

- **SERIAL DATA TRANSMISSION:** is when data is sent, *one bit at a time*, over a single wire or channel (bits are sent one after the other in a single stream).

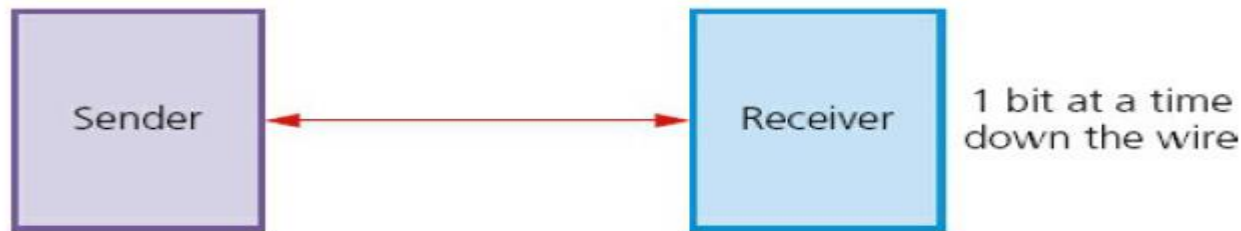


Figure 2.1

(Note: bits can be transmitted as simplex, half-duplex or full-duplex.)

This method of data transmission works well over long distances. However, data is transmitted at a slower rate than parallel data transmission. Since only one wire or channel is used, there is no problem of data arriving at its destination out of synchronisation.

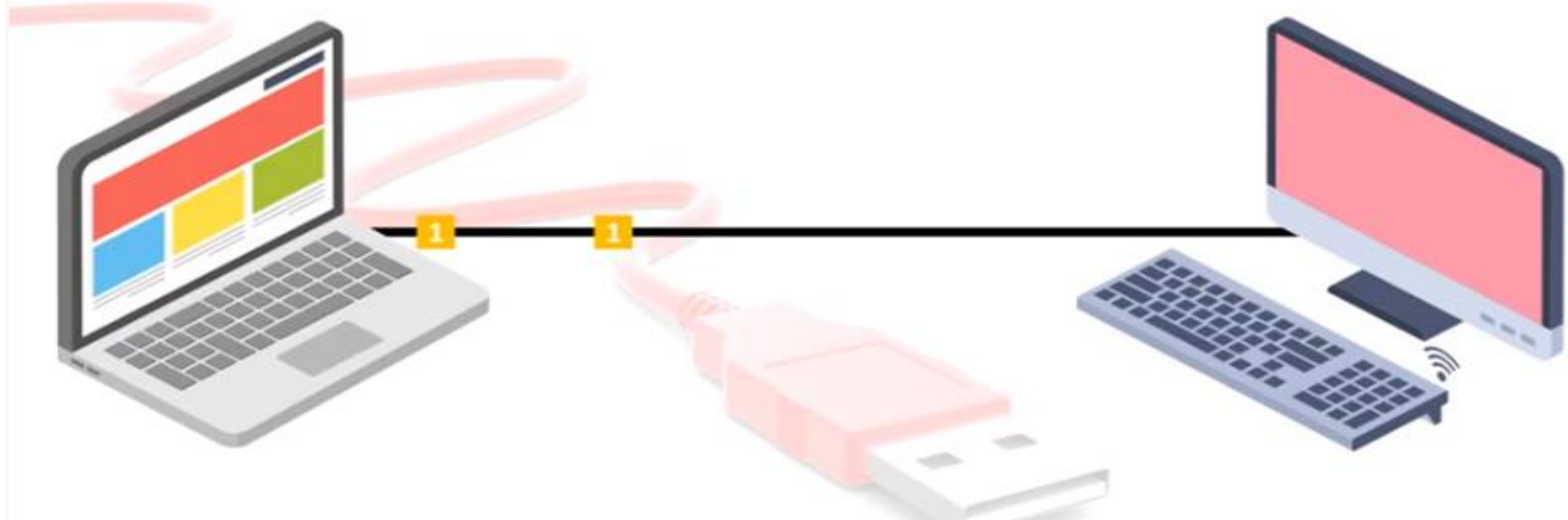
An example of its use is sending data from a computer to a modem for transmission over a telephone line.

2.1 Types and methods of data transmission

2.1.2 Data transmission

Serial data transmission

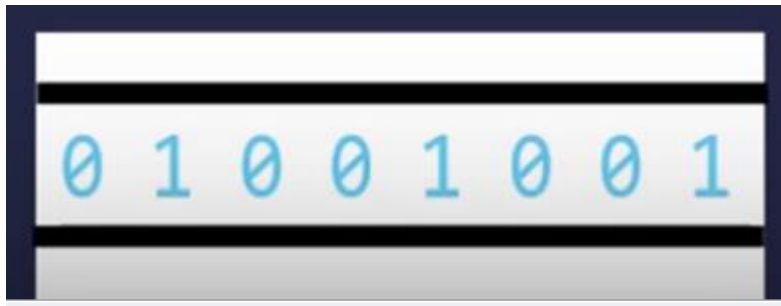
Individual bits of data are sent one at a time via a single cable.



2.1.2 Data transmission

Serial transmission

- When bits is transmitted 1 bit at a time.
- Data is transmitted using a single wire.
- Bits arrive in order.
- Transmit data over long distance with high chance of accuracy.
- Application: **USB / WIFI**

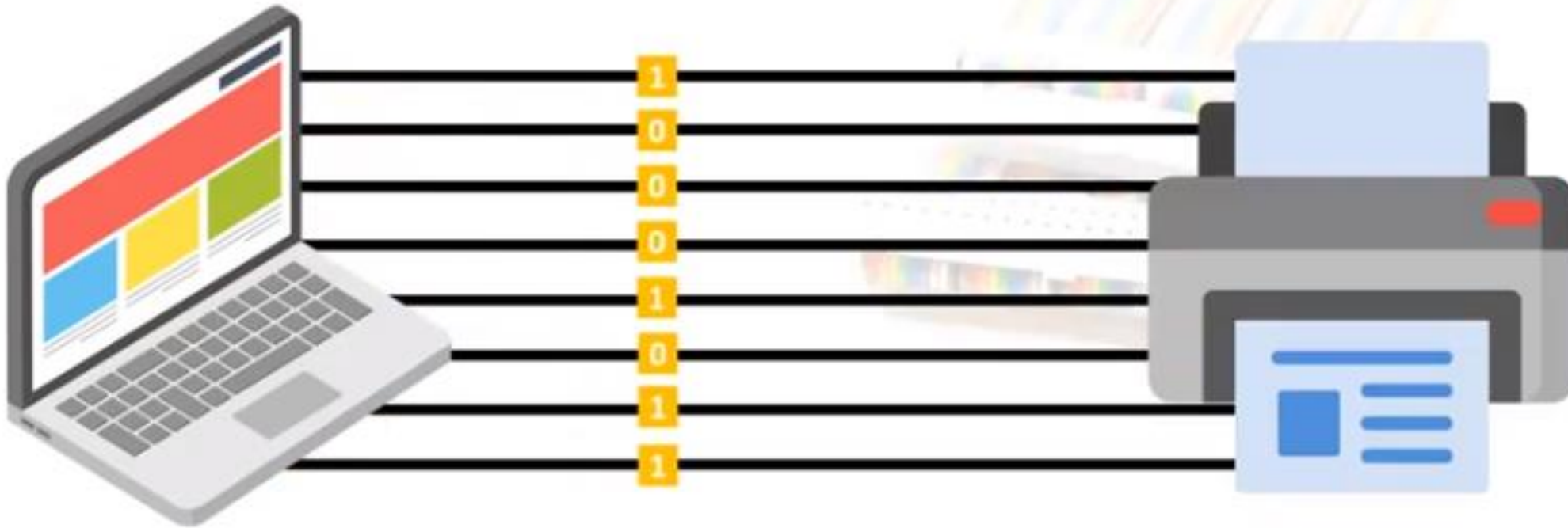


2.1 Types and methods of data transmission

2.1.2 Data transmission

Parallel data transmission

Individual bits of data are sent simultaneously via a series of cables.



2.1.2 Data transmission

- **PARALLEL DATA TRANSMISSION:** is when several bits of data (usually 1 byte) are sent down several wires or channels at the same time; one wire or channel is used to transmit each bit.

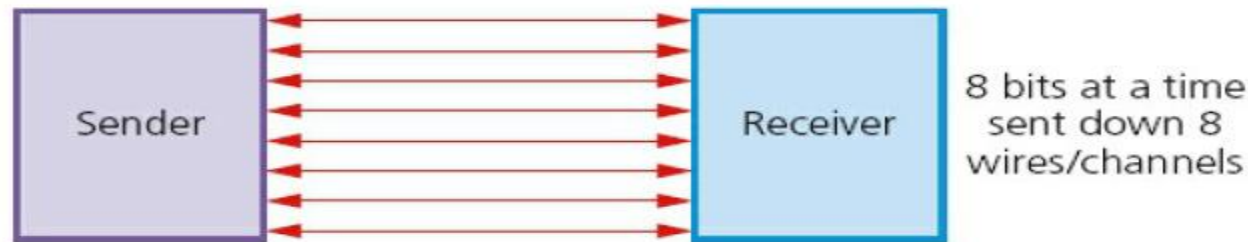


Figure 2.2

(Note: bits can be transmitted as simplex, half-duplex or full-duplex.)

This method of data transmission works very well over short distances (over longer distances, the bits can become 'skewed' (suddenly change direction or position) – this means they will no longer be synchronized (operate at the same time)). It is, however, a faster method of data transmission than serial.

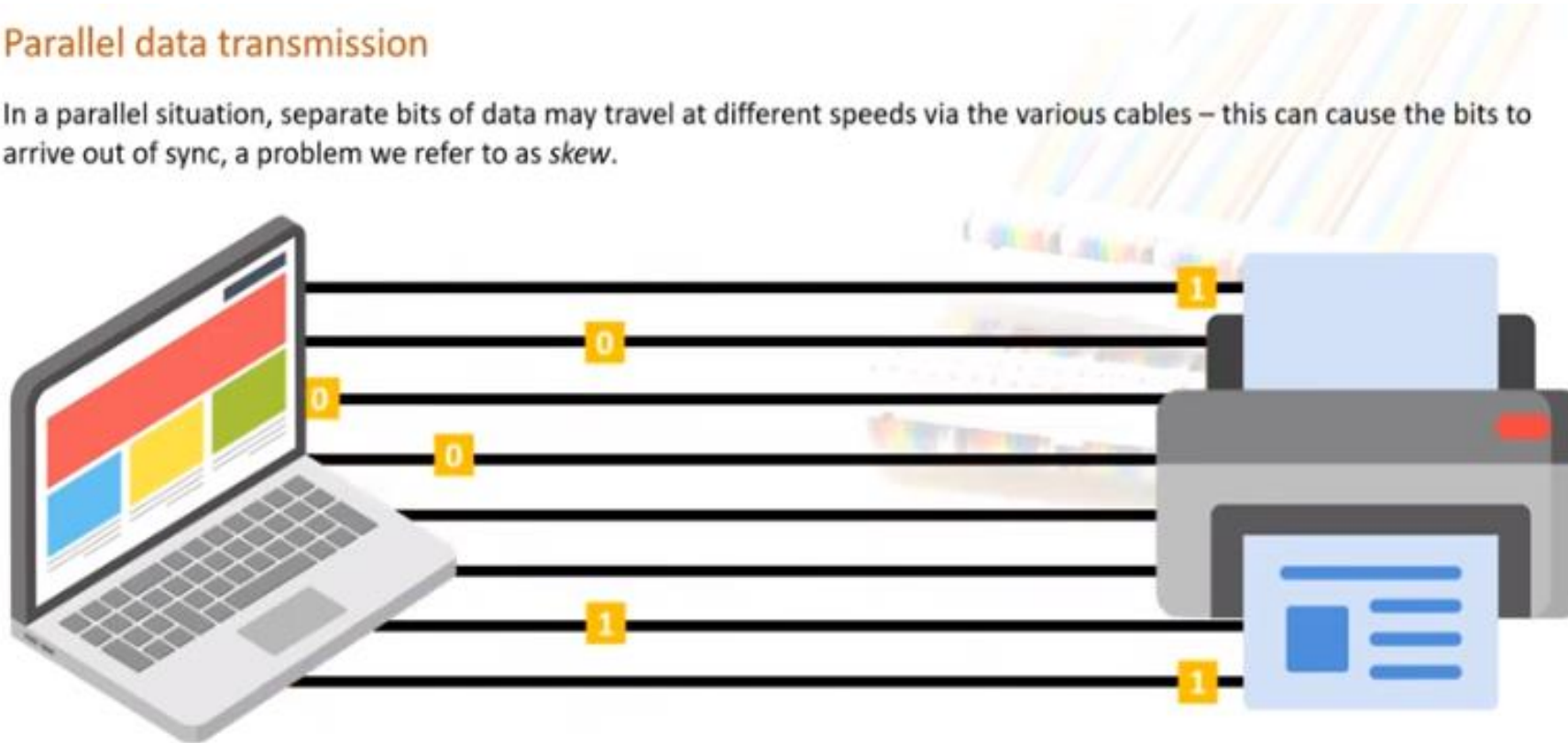
An example of its use is when sending data to a printer from a computer using a ribbon connector.

2.1 Types and methods of data transmission

2.1.2 Data transmission

Parallel data transmission

In a parallel situation, separate bits of data may travel at different speeds via the various cables – this can cause the bits to arrive out of sync, a problem we refer to as *skew*.



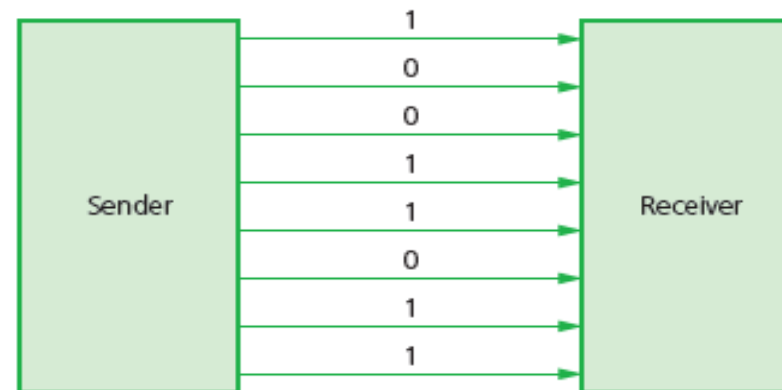
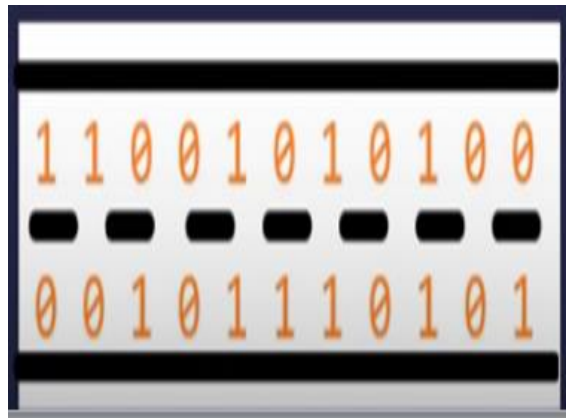
That is why parallel cables are only used over very short distances – and another disadvantage against serial cables.

skewed (Suddenly change direction/position) due to being sent at the same time.

2.1.2 Data transmission

Parallel transmission

- Transmit multiple bits of data all at once.
- Used when a user has a large amount of data to transmit and this need to be done as quickly as possible.



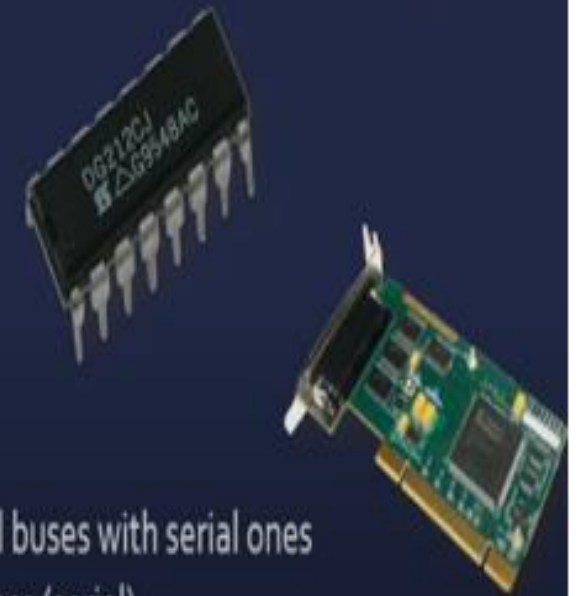
2.1.2 Data transmission

Parallel transmission

- Short range transfer
- Can be used in integrated circuits, ram and connection in peripheral devices such as printers.
- Data is transmitted using multiple wires.
- Bits may arrive out of sequence.

Parallel

- Very short range transfers
- E.g. in ICs (Integrated Circuits), in buses, within memory
- BUT, the trend is to replace parallel buses with serial ones
- E.g. from PCI (parallel) to PCI Express (serial)



2.1 Types and methods of data transmission

2.1.2 Data transmission

Serial vs parallel transmission

Serial transmission		Parallel transmission	
	Commonly used with the USB (Universal Serial Bus) interface		Commonly used with printer cables and other internal device connections
	Minimal interference Simple and low-cost Better reliability over longer distances		Very fast transmission speeds Data can be sent in both directions at the same time
	Lower transmission speeds		Can only be used over very short distances More prone to interference

2.1.2 Data transmission

Parallel over serial

Benefits of parallel over serial:

- Faster data transfer.

Drawback (disadvantage) of parallel over serial:

- More chance of data being skewed (Suddenly change direction/position) due to being sent at the same time.
- More expensive as requires multiple wires.
- More chance of interference as more wires are used.

2.1.2 Data transmission

Advantages of Serial transmission

- It uses a single wire therefore it is cheaper to buy and less likely to have interference which mean it can be used over a longer distances.
- Data is sent a bit at a time so data is received in order.
- Transmission can be synchronised.
- It can reduce rate of errors.

2.1 Types and methods of data transmission

2.1.2 Data transmission



Data transmission methods

The direction that data is allowed to travel via a cable can be described in one of three ways:

- **Simplex:** One direction.
- **Half-duplex:** Both directions, but not at the same time.
- **Full-duplex:** Both directions at the same time.

Along with the direction of data transmission, data can also be sent via **serial** or **parallel** transmission.

- **Serial transmission:** Data bits are sent one at a time via a single cable.
 - Lower interference than parallel.
 - More reliable over longer distances.
 - Cheaper to set up.
- **Parallel transmission:** Several bits of data can be sent simultaneously via a series of cables.
 - More prone to interference.
 - Data may arrive out of order – known as *skew*.
 - Limited to very short distances.
 - Very fast data transfer.

2.1 Types and methods of data transmission

<https://www.youtube.com/watch?v=Pe4dIlmwDhY>

2.1.3 Universal Serial Bus (USB)



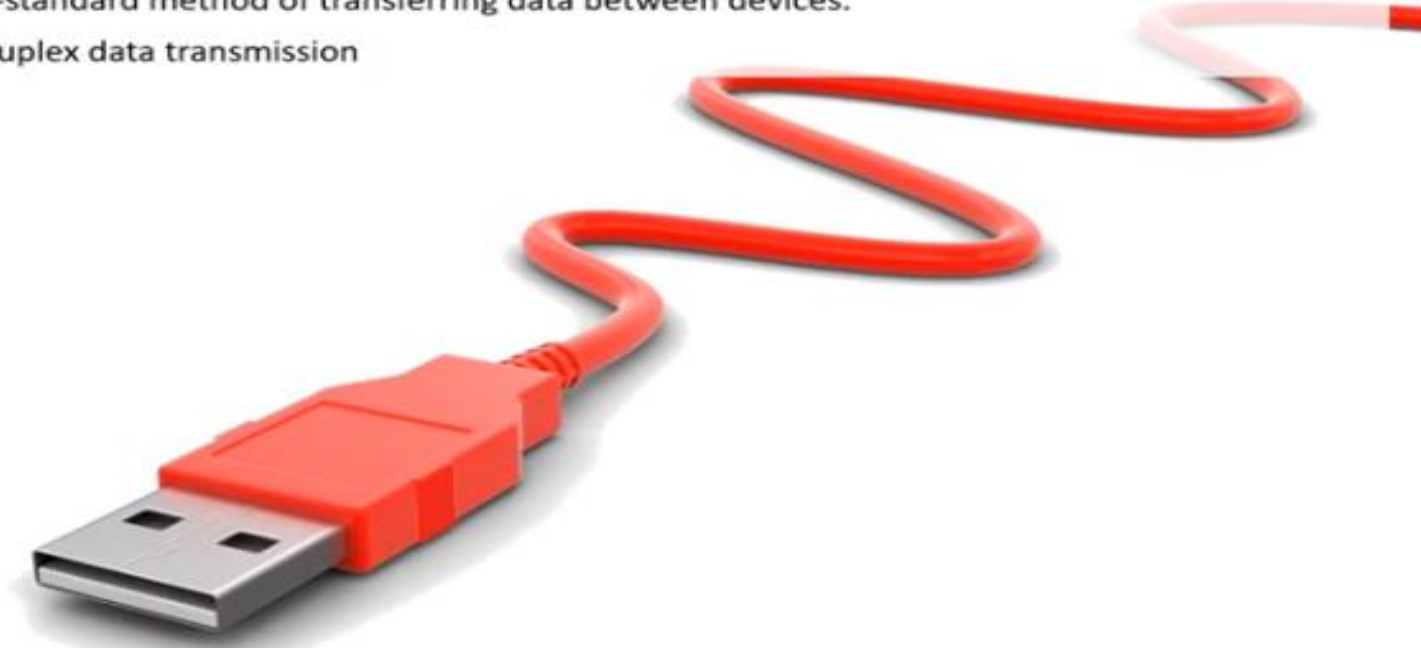
2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

Universal Serial Bus

Universal Serial Bus (USB) is a serial method of transmitting data. It has become the most common type of input/output used in computing and an industry-standard method of transferring data between devices.

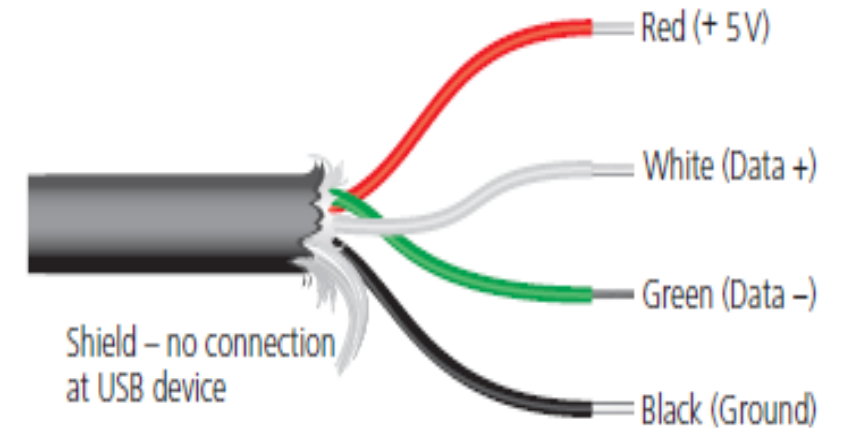
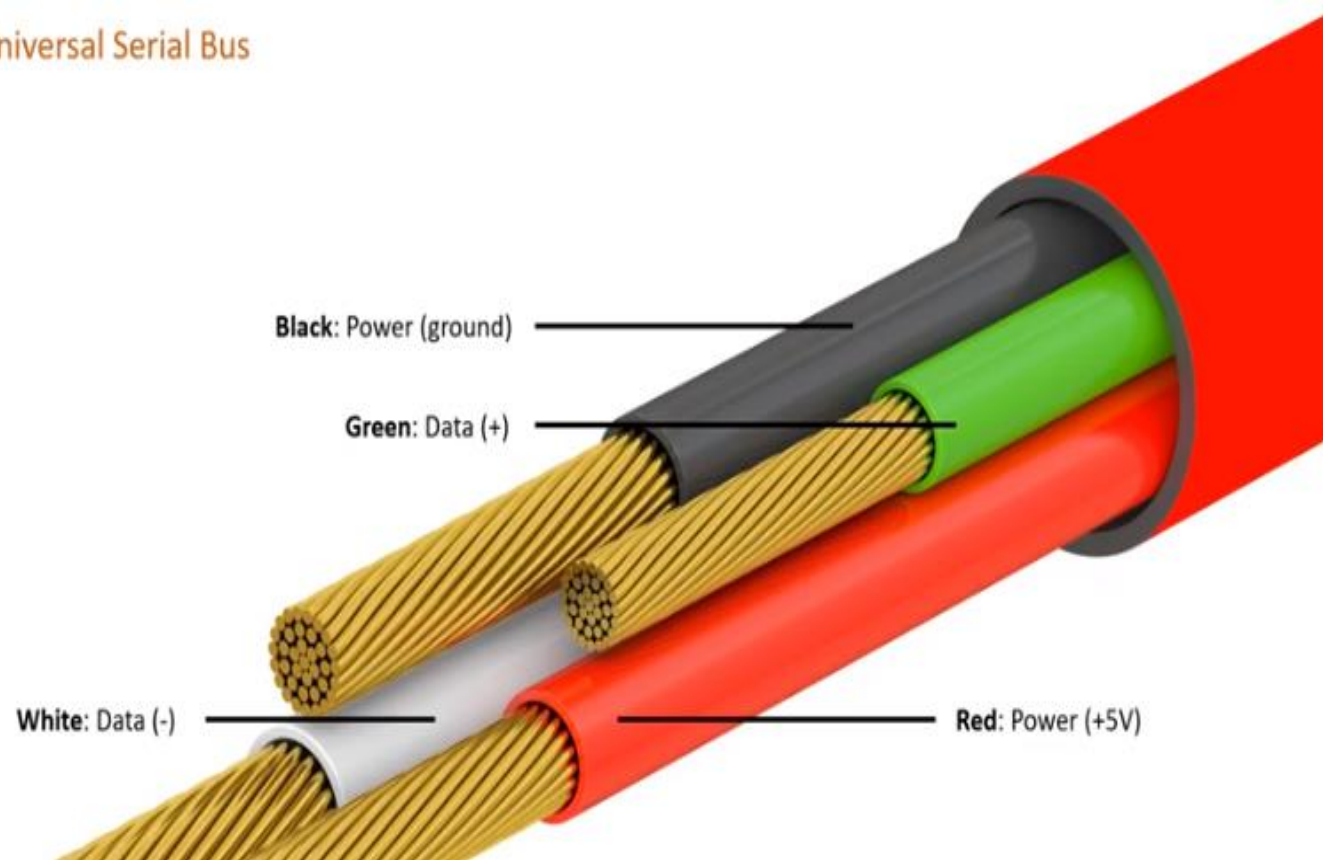
USB allows for both half- and full-duplex data transmission



2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

Universal Serial Bus



2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

Benefits of USB

-  Industry-standard – and therefore, very well-supported.
-  Devices connected via USB are automatically detected and appropriate device drivers loaded.
-  Supplies +5V of power, so most USB-connected devices do not need an additional power source.
-  Designed to be backwards-compatible
-  A USB hub can be used to easily add extra ports to an existing device.
-  USB protocol can notify the transmitting device to re-transmit data if it detects errors.

2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

Drawbacks of USB



Limited distance

- Only supports cable lengths of around 5m.
- Can be extended but only with additional hardware.



Limitations of backwards-compatibility

- Support for older standards is becoming increasingly rare.



Data transfer limitations

- Even newer versions like USB 3.0 and USB-C have slower data transfer speeds compared to Ethernet.

2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

- **The UNIVERSAL SERIAL BUS (USB)** is an **asynchronous** serial data transmission method. It has quickly become the standard method for transferring data between a computer and a number of devices. Essentially, the USB cable consists of:
 - a four-wire shielded cable
 - two of the wires are used for power and the earth
 - two of the wires are used in the data transmission.



2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

Benefits

Disadvantage

✓	✗
Devices plugged into the computer are automatically detected; device drivers are automatically uploaded	—
The connectors can only fit one way; this prevents incorrect connections being made	The maximum cable length is presently about 5 metres
This has become the industry standard; this means that considerable support is available to users	—
Several different data transmission rates are supported	The present transmission rate is limited to less than 500 megabits per second
Newer USB standards are backward compatible with older USB standards	The older USB standard (e.g. 1.1) may not be supported in the near future

2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

USB-C



USB Type A

USB Type B

USB 3.0

USB Mini

USB Micro

USB Type C

USB Micro B

2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)

USB-C

USB-C is becoming more and more common in laptops, tablets and smartphones.

Unlike previous versions, USB-C uses a symmetrical 24-pin connector that fits into the port either way round.

Advances

- Smaller and thinner than previous versions.
- Offers 100 watts (20 volts) of power – full-sized devices can now be charged via USB.
- Supports data transfer of up to 10 gigabits per second – enough for 4K video streaming.
- Backwards-compatibility with USB 2.0 and 3.0



2.1 Types and methods of data transmission

2.1.3 Universal Serial Bus (USB)



Universal Serial Bus

Universal Serial Bus (USB) is a form of serial data transmission supporting both half- and full-duplex.

Benefits

- Very well-supported.
- Connected devices are automatically detected and device drivers loaded.
- Most USB-connected devices do not need an additional power source.
- Designed to be backwards-compatible.
- A USB hub can be used to easily add extra ports to an existing device.
- USB protocol can notify the transmitting device to re-transmit data if it detects errors.

Drawbacks

- Limited distance of around 5m.
- Many devices no longer support older standards.
- Newer versions have slower data transfer speeds compared to Ethernet.

IGCSE Past Exam Paper Question 1:

(c) Julia uses a USB connection to transfer data onto her USB flash memory drive.

(i) One benefit of using a USB connection is that it is a universal connection.

State **two** other benefits of using a USB connection.

Benefit 1

.....

Benefit 2

.....

[2]

(ii) Identify the type of data transmission used in a USB connection.

..... [1]

IGCSE Past Exam Paper Answer 1:

2(c)(i)	Any two from: – It cannot be inserted incorrectly – Supports different transmission speeds – High speed transmission – Automatically detected (not connected) // automatically downloads drivers – Powers the device (for data transfer) – Backward compatible	2
2(c)(ii)	– Serial	1

IGCSE Past Exam Paper Question 2:

2 A school network is used to transmit and store data about students.

(a) Different types and methods of transmission can be used to send data across the network.

Three descriptions about data transmission are given.

Tick (✓) **one Method** and tick (✓) **one Type** for each description.

Description	Method		Type		
	Serial (✓)	Parallel (✓)	Simplex (✓)	Half- duplex (✓)	Duplex (✓)
Data is sent down a single wire in a single direction only.					
Data is sent down multiple wires in both directions, at the same time.					
Data is sent down a single wire in both directions, but never at the same time.					

[3]

IGCSE Past Exam Paper Answer 2:

Question	Answer	Mark																													
2(a)	1 mark per each correct row: <table><tr><th rowspan="2">Description</th><th colspan="2">Method</th><th colspan="3">Type</th></tr><tr><th>Serial (✓)</th><th>Parallel (✓)</th><th>Simplex (✓)</th><th>Half- duplex (✓)</th><th>Duplex (✓)</th></tr><tr><td>Data is sent down a single wire in a single direction only.</td><td>✓</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>Data is sent down multiple wires in both directions, at the same time.</td><td></td><td>✓</td><td></td><td></td><td>✓</td></tr><tr><td>Data is sent down a single wire in both directions, but never at the same time.</td><td>✓</td><td></td><td></td><td>✓</td><td></td></tr></table>	Description	Method		Type			Serial (✓)	Parallel (✓)	Simplex (✓)	Half- duplex (✓)	Duplex (✓)	Data is sent down a single wire in a single direction only.	✓		✓			Data is sent down multiple wires in both directions, at the same time.		✓			✓	Data is sent down a single wire in both directions, but never at the same time.	✓			✓		3
Description	Method		Type																												
	Serial (✓)	Parallel (✓)	Simplex (✓)	Half- duplex (✓)	Duplex (✓)																										
Data is sent down a single wire in a single direction only.	✓		✓																												
Data is sent down multiple wires in both directions, at the same time.		✓			✓																										
Data is sent down a single wire in both directions, but never at the same time.	✓			✓																											

IGCSE Past Exam Paper Question 3:

- 8 Leonard has a new laser printer to print letters for his business.

Leonard connects his printer to his computer using the USB port.

- (a) Give **three** benefits of using the USB port to connect the printer to the computer.

Benefit 1

.....

Benefit 2

.....

Benefit 3

.....

[3]

IGCSE Past Exam Paper Answer 3:

Question	Answer	Marks
8(a)	Any three from: - It is a universal standard - It can't be inserted the wrong way around - Supports different transmission speeds - Automatically detects if correct driver installed	3

IGCSE Past Exam Paper Question 4:

(c) Priya stores her website on a webserver.

To transmit the website data to the webserver she uses parallel duplex data transmission.

Describe how data is transmitted using parallel duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

IGCSE Past Exam Paper Answer 4:

PUBLISHED

Question	Answer	Marks
6(c)	- Several/multiple bits are transmitted at a time/simultaneously - Several/multiple wires are used - Data is transmitted in both directions ... - ... at the same time/simultaneously	4

IGCSE Past Exam Paper Question 5:

7 Computers can use different methods of transmission to send data from one computer to another.

Parallel data transmission is one method that can be used.

(a) Explain what is meant by parallel data transmission.

.....
.....
.....
..... [2]

(b) Give **one** benefit and **one** drawback of parallel data transmission, compared to serial data transmission, over short distances.

Benefit
.....
.....
Drawback
.....
..... [2]

(c) Give **one** example where parallel data transmission is used.

..... [1]

IGCSE Past Exam Paper Answer 5:

PUBLISHED

Question	Answer	Marks
7(a)	1 mark for each correct answer: ☐ uses several/multiple wires ☐ transmits multiple bits at a time	2
7(b)	Benefit 1 mark for: ☐ quicker/faster data transfer Drawback One from: ☐ More chance of data being skewed due to bits being sent simultaneously/out of order // less safe transmission as bits are sent simultaneously/out of order ☐ More expensive as requires more/several/multiple wires ☐ More chance of interference as more/several/multiple wires are used (than can create crosstalk)	2
7(c)	One from: ☐ Used in integrated circuits ☐ Used in RAM ☐ Used in connections to peripheral devices (e.g. printer)	1

IGCSE Past Exam Paper Question 6:

9 (a) Computers can transmit data using different methods.

Describe the **three** data transmission methods given.

(i) Serial data transmission

.....
.....
.....
.....[2]

(ii) Parallel data transmission

.....
.....
.....
.....[2]

(iii) Duplex data transmission

.....
.....
.....
.....[2]

IGCSE Past Exam Paper Answer 6:

Question	Answer	Marks
9(a)(i)	Two from: ☐ Data is transmitted one bit at a time ☐ Data is transmitted using a single wire ☐ Bits arrive in order/sequence	2
9(a)(ii)	Two from: ☐ Data is transmitted multiple bits at a time/simultaneously ☐ Data is transmitted using multiple wires ☐ Bits may arrive out of sequence/skewed (and are reordered)	2
9(a)(iii)	1 mark for each: ☐ Data is transmitted in both directions ☐ ... at the same time/simultaneously	2

IGCSE Past Exam Paper Question 7:

5 Data can be transferred using half-duplex serial transmission.

(a) Describe serial transmission.

.....
.....
.....
.....[2]

(b) Give **one** application of serial data transmission.

.....
.....[1]

(c) Describe half-duplex data transmission.

.....
.....
.....
.....[2]

IGCSE Past Exam Paper Answer 7:

Question	Answer	Marks
5(a)	☐ Bits sent one at a time ☐ Uses a single wire	2
5(b)	USB / SATA / Wifi / PCI <u>Express</u> / Any appropriate serial device	1
5(c)	☐ Data is transferred in two directions ☐ Data is sent in only one direction at a time	2

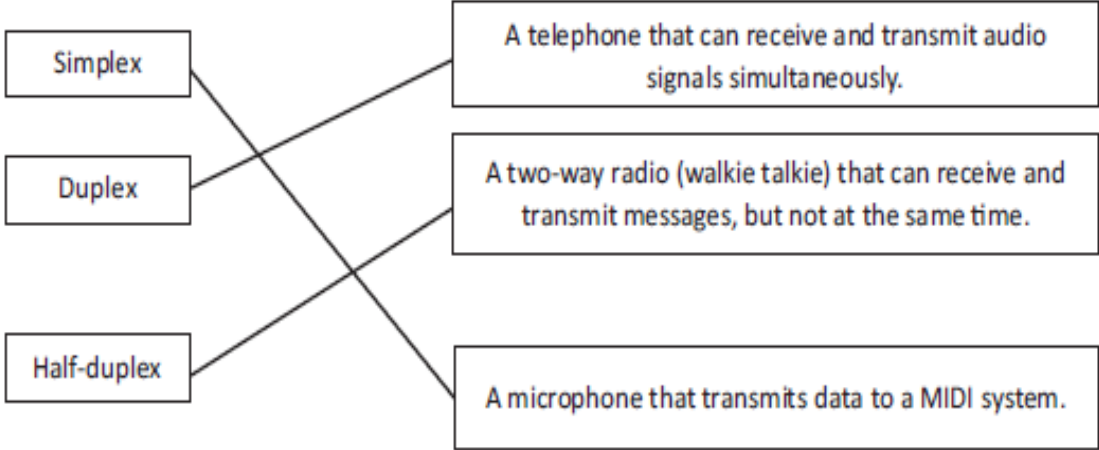
IGCSE Past Exam Paper Question 8:

2 Draw a line to connect each term to the correct application.

Term	Application
Simplex	A telephone that can receive and transmit audio signals simultaneously.
Duplex	A two-way radio (walkie-talkie) that can receive and transmit messages, but not at the same time.
Half-duplex	A microphone that transmits data to a MIDI system.

[2]

IGCSE Past Exam Paper Answer 8:

Question	Answer		Marks
2	Term	Application	2
			
Three correct lines = 2 marks Two or one correct line = 1 mark			

IGCSE Past Exam Paper Question 9:

- 4 Five statements about **serial half-duplex** data transmission are shown in the table below.

Tick (✓) to show whether each statement is **true** or **false**.

Statement	true (✓)	false (✓)
Data is transmitted in one direction only, one bit at a time.		
Data is transmitted in both directions, multiple bits at a time.		
Data is transmitted in one direction only, multiple bits at a time.		
Data is transmitted in both directions, but only one direction at a time. Data is transmitted one bit at a time.		
Data is transmitted in both directions, but only one direction at a time. Data is transmitted multiple bits at a time.		

[5]

IGCSE Past Exam Paper Answer 9:

Question	Answer	Marks																		
4	1 mark per correct tick <table><tr><th>Statement</th><th>True</th><th>False</th></tr><tr><td>Data is transmitted in one direction only, one bit at a time.</td><td></td><td>✓</td></tr><tr><td>Data is transmitted in both directions, multiple bits at a time.</td><td></td><td>✓</td></tr><tr><td>Data is transmitted in one direction only, multiple bits at a time.</td><td></td><td>✓</td></tr><tr><td>Data is transmitted in both directions, but only one direction at a time. Data is transmitted one bit at a time.</td><td>✓</td><td></td></tr><tr><td>Data is transmitted in both directions, but only one direction at a time. Data is transmitted multiple bits at a time.</td><td></td><td>✓</td></tr></table>	Statement	True	False	Data is transmitted in one direction only, one bit at a time.		✓	Data is transmitted in both directions, multiple bits at a time.		✓	Data is transmitted in one direction only, multiple bits at a time.		✓	Data is transmitted in both directions, but only one direction at a time. Data is transmitted one bit at a time.	✓		Data is transmitted in both directions, but only one direction at a time. Data is transmitted multiple bits at a time.		✓	5
Statement	True	False																		
Data is transmitted in one direction only, one bit at a time.		✓																		
Data is transmitted in both directions, multiple bits at a time.		✓																		
Data is transmitted in one direction only, multiple bits at a time.		✓																		
Data is transmitted in both directions, but only one direction at a time. Data is transmitted one bit at a time.	✓																			
Data is transmitted in both directions, but only one direction at a time. Data is transmitted multiple bits at a time.		✓																		

IGCSE Past Exam Paper Question 10:

7 Computer A is communicating with computer B.

- (a) Draw an arrow or arrows to show simplex, duplex and half-duplex data transmission. The **direction** of the data transmission must be fully **labelled**.

Simplex data transmission



Computer A



Computer B

Duplex data transmission



Computer A



Computer B

Half-duplex data transmission



Computer A



Computer B

[6]

IGCSE Past Exam Paper Question 10:

- (b)** State a use for the following data transmission methods. The use must be different for each data transmission method.

Simplex

.....

Duplex

.....

[2]

- (c)** A computer includes an Integrated Circuit (IC) and a Universal Serial Bus (USB) for data transmission.

Describe how the computer uses these for data transmission, including the type of data transmission used.

IC

.....

.....

.....

USB

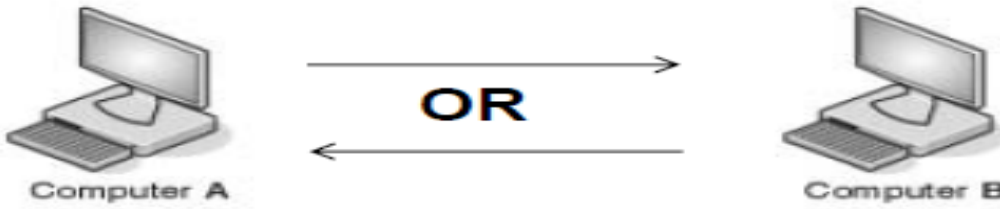
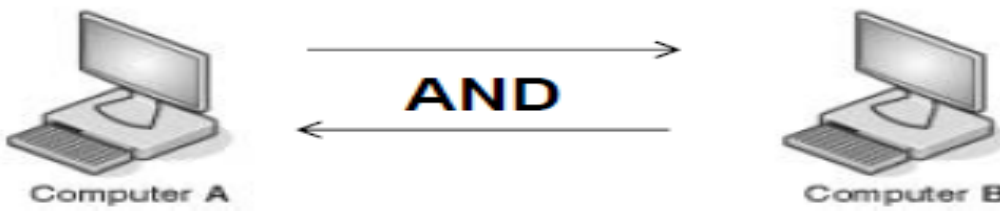
.....

.....

.....

[4]

IGCSE Past Exam Paper Answer 10:

Question	Answer	Marks
7(a)	1 mark for correct arrow(s), one mark for correct description Simplex data transmission  Computer A Computer B (Direction of data is) one way only // unidirectional Duplex data transmission  Computer A Computer B (Direction of data is both ways) <u>at same time</u> / <u>simultaneously</u> / <u>concurrently</u> Half-duplex data transmission  Computer A Computer B	6

IGCSE Past Exam Paper Answer 10:

7(b)	1 mark each use, must be different. Simplex e.g.: Microphone to computer Sensor to computer Computer to printer Computer to speaker Computer to monitor Webcam to computer Sending data to a device // sending data from a device Duplex e.g.: Telephone call Voice over IP Computer to printer (only award once) Instant messaging Broadband connections Video conferencing Sending data to and from devices e.g wireless technology Computer to modem	2
7(c)	2 marks for IC, 2 marks for USB IC ☐ parallel transmission // description of parallel ☐ for sending data internally USB ☐ serial transmission // description of serial ☐ for sending data externally (to and from peripherals / between devices)	4

IGCSE Past Exam Paper Question 11:

4 (a) Nikita wishes to print out some documents and connects her printer to the computer using one of the USB ports.

(i) Identify what type of data transmission is being used.

.....[1]

(ii) Give **three** reasons for using a USB port.

1

.....

2

.....

3

.....

[3]

IGCSE Past Exam Paper Answer 11:

4 (a) (i) serial [1]

(ii) Any **three** from:

- automatically detects the hardware/installs drivers
- plug only goes in one way/can't connect incorrectly
- supports different data transmission speeds/a range of data transmission speeds
- has become the industry standard/universally used
- backwards compatible (with earlier versions of USB ports)

[3]

IGCSE Past Exam Paper Question 12:

6 (a) Three descriptions of data transmission are given below.

Tick (✓) the appropriate box in each table to show the:

- type of transmission
- method of transmission

Description 1:

Data is transmitted several bits at a time down several wires in both directions simultaneously.

Type	Tick (✓)
simplex	
half-duplex	
full-duplex	

Method	Tick (✓)
serial	
parallel	

Description 2:

Data is transmitted in one direction only, one bit at a time, down a single wire.

Type	Tick (✓)
simplex	
half-duplex	
full-duplex	

Method	Tick (✓)
serial	
parallel	

Description 3:

Data is transmitted one bit at a time down a single wire; the data is transmitted in both directions but not at the same time.

Type	Tick (✓)
simplex	
half-duplex	
full-duplex	

Method	Tick (✓)
serial	
parallel	

[6]

IGCSE Past Exam Paper Question 12:

- (b) Give **two** reasons why serial transmission, rather than parallel transmission, is used to connect devices to a computer.

1

.....

2

.....

[2]

IGCSE Past Exam Paper Answer 12:

6 (a)

Type	Tick (✓)
simplex	
half-duplex	
full-duplex	✓

Method	Tick (✓)
serial	
parallel	✓

Type	Tick (✓)
simplex	✓
half-duplex	
full-duplex	

Method	Tick (✓)
serial	✓
parallel	

Type	Tick (✓)
simplex	
half-duplex	✓
full-duplex	

Method	Tick (✓)
serial	✓
parallel	

[6]

IGCSE Past Exam Paper Answer 12:

(b) Any two from:

- single wire means there is less chance of interference/data corruption
- single wire reduces costs
- more reliable over greater distances
- bits will still be synchronised after transmission

[2]

Learning Objectives

2.2 Methods of error detection

Candidates should be able to:

- 1 Understand the need to check for errors after data transmission and how these errors can occur
- 2 Describe the processes involved in each of the following error detection methods for detecting errors in data after transmission: parity check (odd and even), checksum and echo check
- 3 Describe how a check digit is used to detect errors in data entry and identify examples of when a check digit is used, including international standard book numbers (ISBN) and bar codes
- 4 Describe how an automatic repeat query (ARQ) can be used to establish that data is received without error

Notes and guidance

- Errors can occur during data transmission due to interference, e.g. data loss, data gain and data change
- Including parity byte and parity block check
- Including the use of:
 - positive/negative acknowledgements
 - timeout

2.2 Methods of error detection

https://www.youtube.com/watch?v=ckvEN_JyiKE

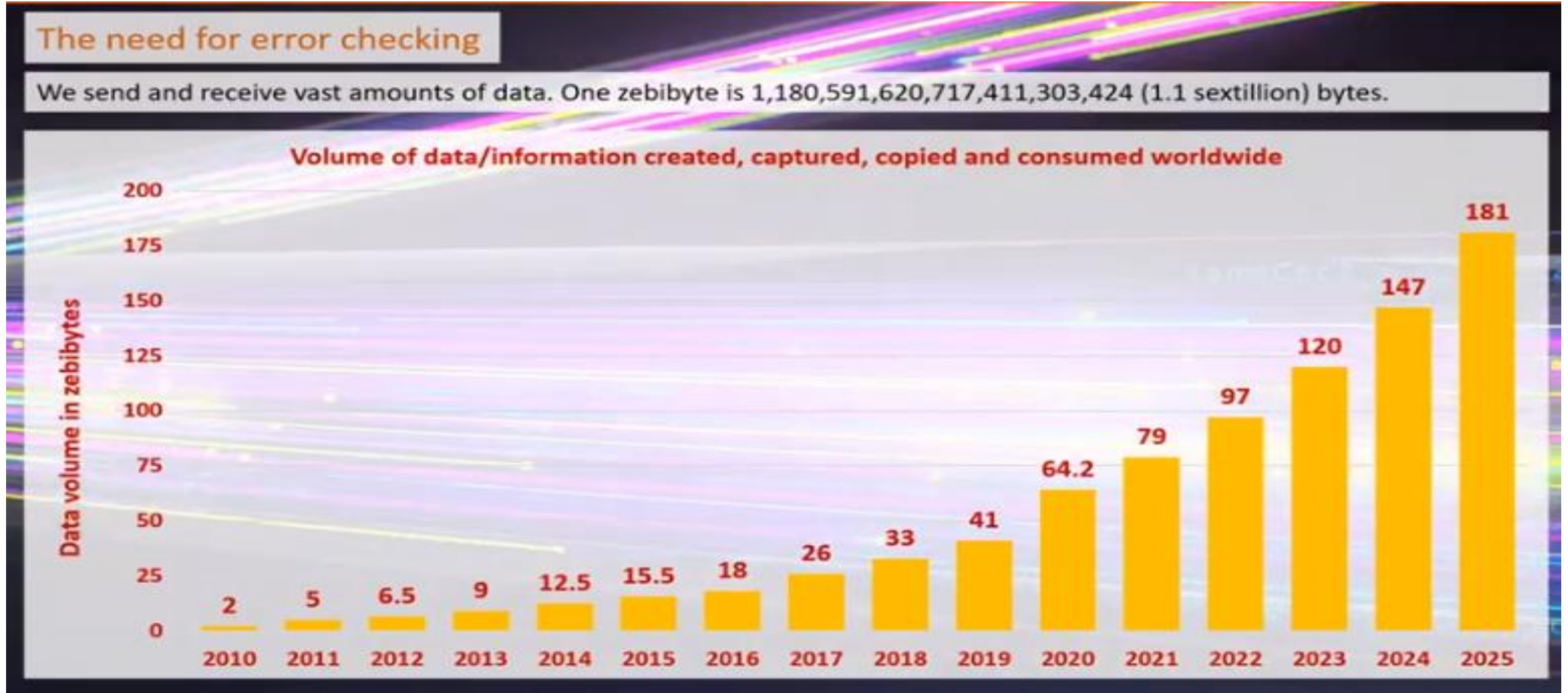
2.2 Methods of error detection

The need for error checking



2.2 Methods of error detection

The need to check for errors



2.2 Methods of error detection

The need to check for errors

The need for error checking

With the complexity of computers and the vast amounts of 1s and 0s being transmitted, there is always a possibility that data can be lost, gained or changed along the way.



Interference

Every type of cable can suffer from electrical interference, which can cause data to become lost or corrupt – i.e., 1s become 0s and 0s become 1s.



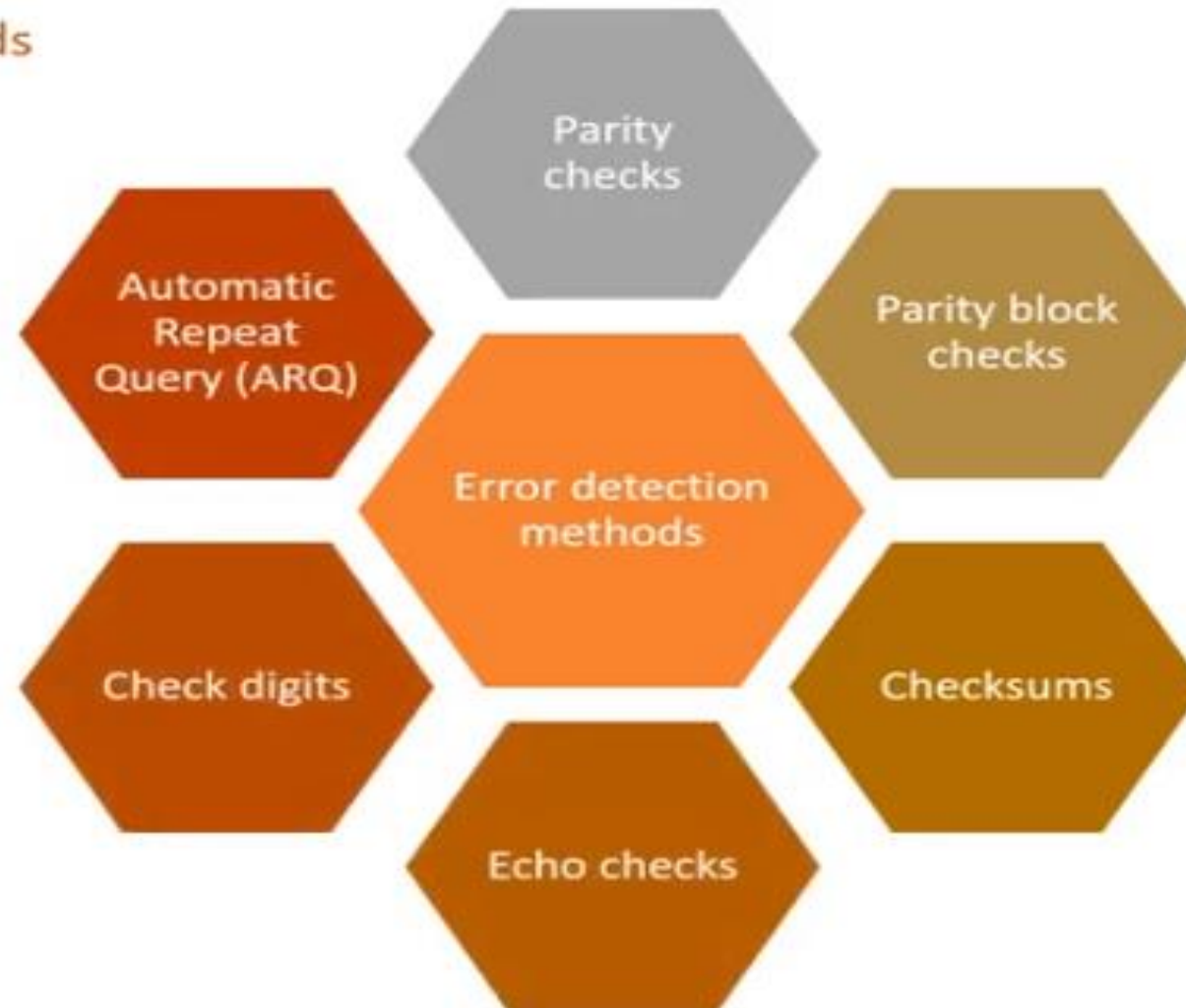
Packet switching

Issues during the packet switching process can cause data to be lost – or in some cases, gained.

2.2 Methods of error detection

The need to check for errors

Error checking methods



2.2 Methods of error detection

The need to check for errors



The need for error checking

When data is transmitted, there is always a possibility that data can be lost, gained or changed.

The most common reasons for data corruption are:

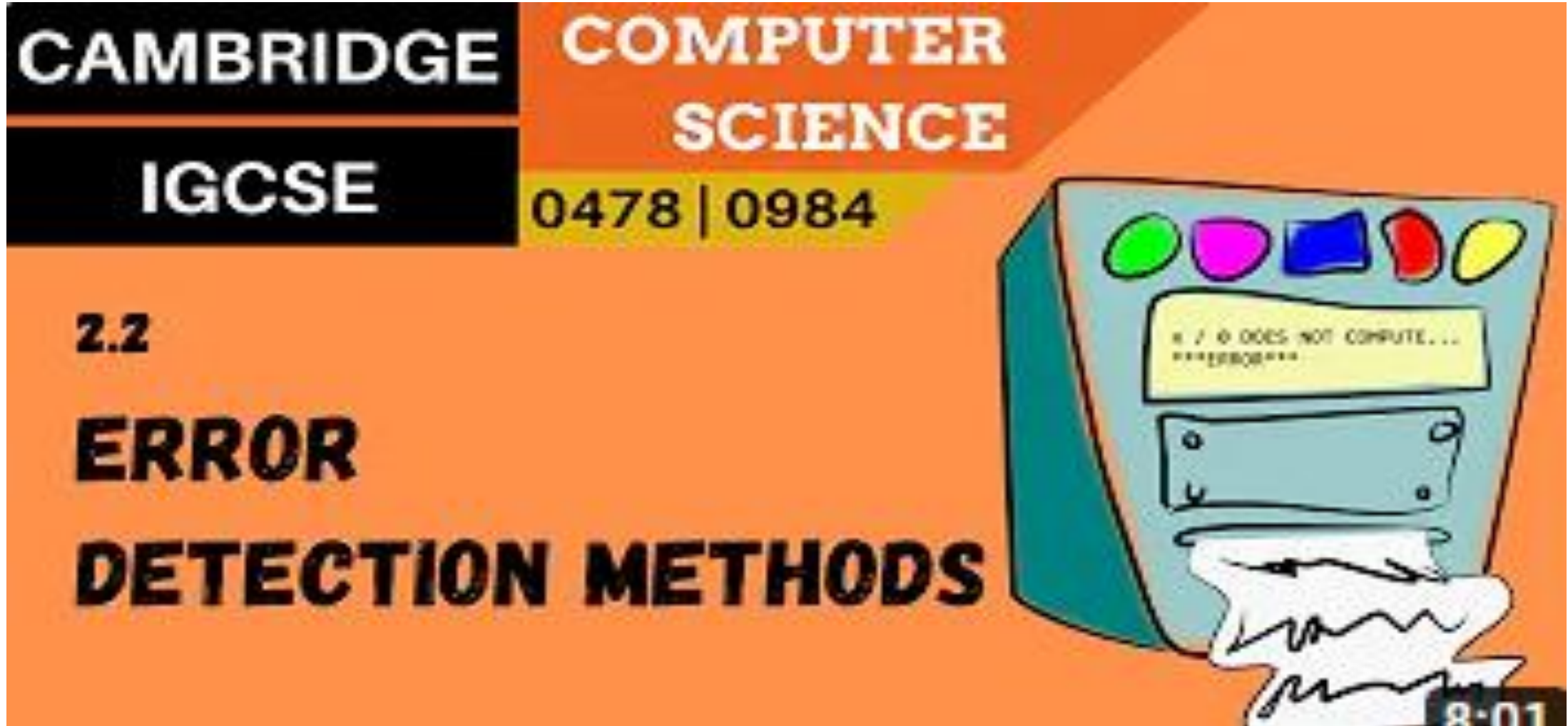
- Electrical interference (noise)
- Problems during packet switching

Common methods of error checking include:

- Parity checks
- Checksums
- Echo checks
- Automatic Repeat Query (ARQ)

2.2 Methods of error detection

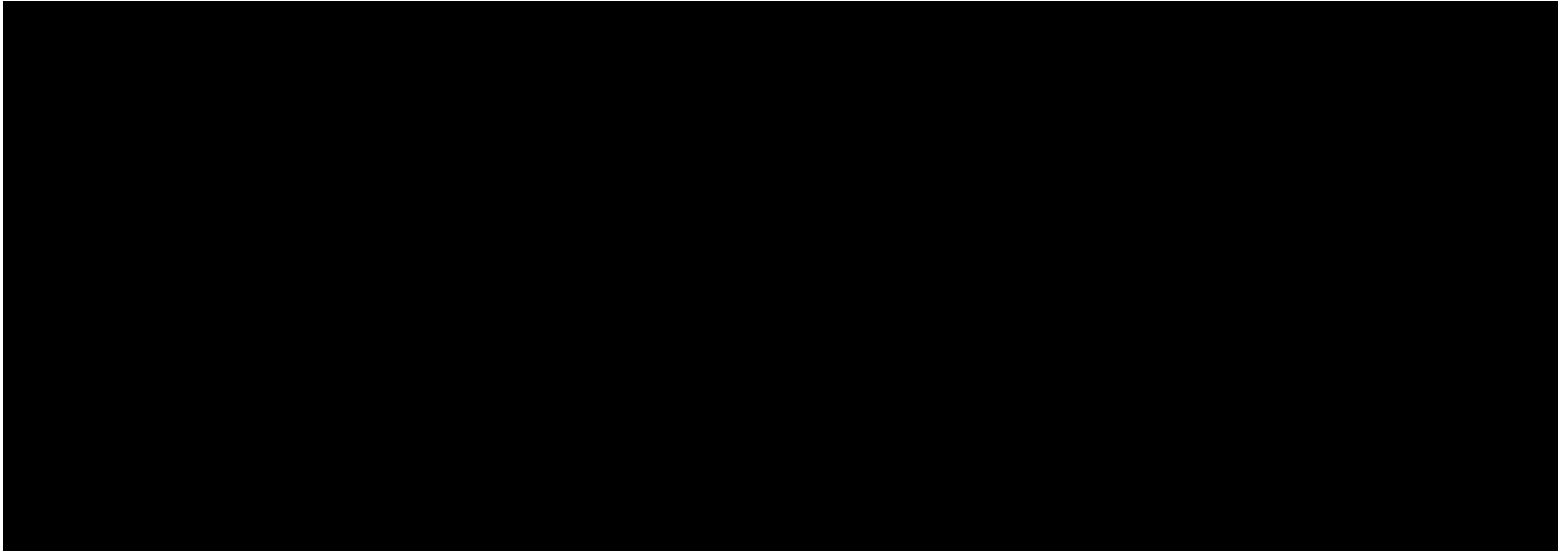
<https://www.youtube.com/watch?v=TEICs3IaKF4>



2.2 Methods of error detection

- Watch the video for Check Digits, Checksums, ARQs and Echo Checks

<https://www.youtube.com/watch?v=AdNcGXejJM4>



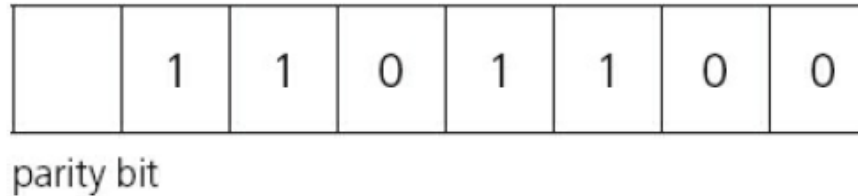
2.2 Methods of error detection

- Following data transmission, there is always the risk that the data has been corrupted or changed in some way. This can occur whether data is being transmitted over short distances or over long distances.
- Checking for errors is important since computers aren't able to check that text is correct; they can only recognize whether a word is in their built-in dictionary or not.
- Look at the following text: **'I cnduo't bvleiee taht I culod aulacly uesdtannrd waht rm onurrddeine was asynig'.**
- A number of methods exist which can detect errors and, in some cases, actually correct the error. The methods covered in this section are:
 - parity checking
 - automatic repeat request (ARQ)
 - checksum
 - echo checking.

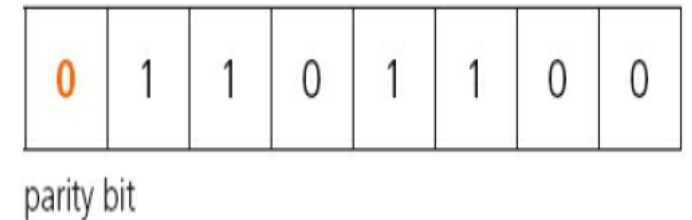
2.2 Methods of error detection

2.2.1. Parity checking

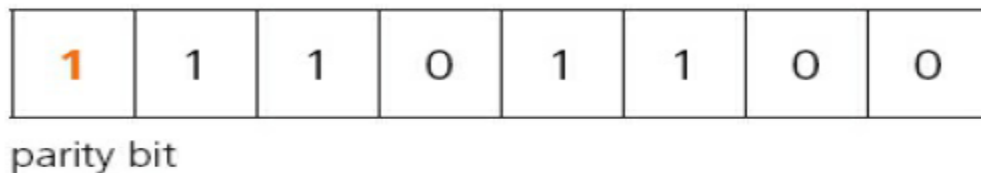
- **PARITY CHECKING** is one method used to check whether data has been changed or corrupted following transmission from one device or medium to another device or medium.
- A byte of data, for example, is allocated a **PARITY BIT**. This is allocated before transmission takes place. Systems that use **EVEN PARITY** have an even number of 1- bits; systems that use **ODD PARITY** have an odd number of 1-bits.
- Consider the following byte:



Therefore, the byte just before transmission would be : either (**even parity**)



or (**odd parity**)



2.2 Methods of error detection

Parity checks (odd and even)

Computers submit data in bytes. As we know, a byte is 8 bits.

In this method, the **Most Significant Bit (MSB)** is reserved and used as a **parity bit**.

We can use either **odd** or **even** parity.

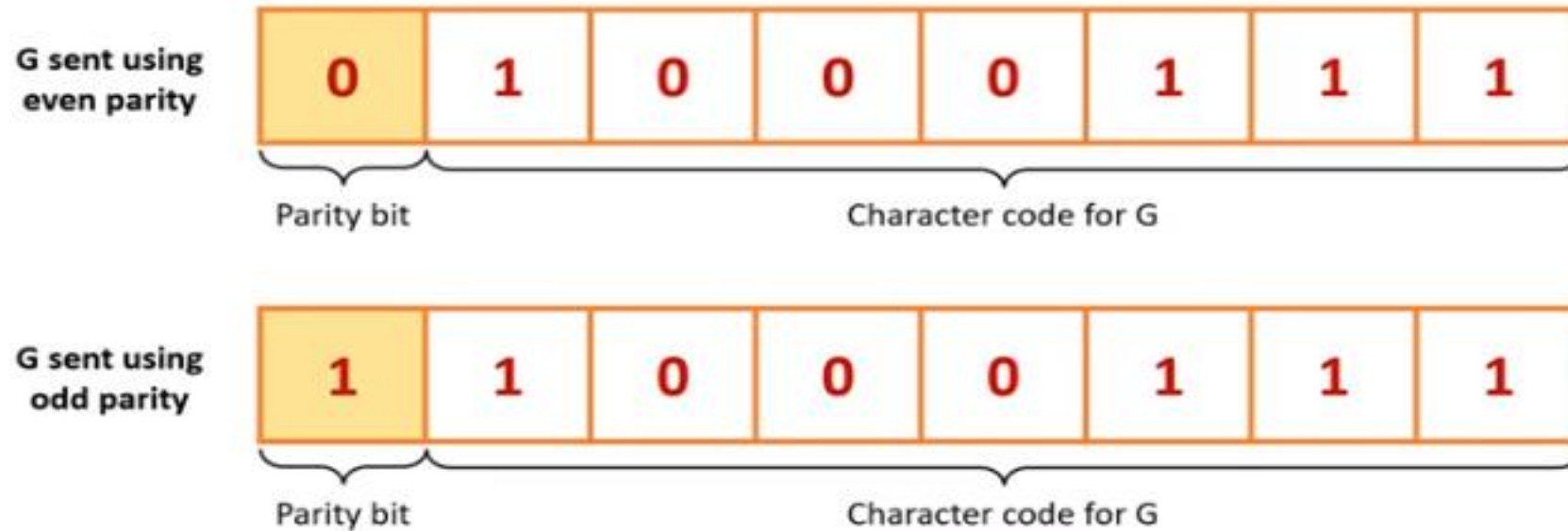
The **parity bit** is set to either 0 or 1 so that the total number of 1s in the byte is either odd or even.

		0	1	0	0	1	0	0
	Parity bit		Data					
Odd parity	1	0	1	0	0	1	0	0
Even parity	0	0	1	0	0	1	0	0

2.2 Methods of error detection

Parity checks (odd and even)

Standard ASCII characters store letters in 7 bits and do not make use of the **Most Significant Bit (MSB)**, so it can potentially be turned into a **parity bit**.



2.2 Methods of error detection

Parity checks (odd and even)

A byte of data being sent using odd parity without an error.



Parity bit

Data



A byte of data being sent using odd parity with a single error.



Parity bit

Data



A byte of data being sent using odd parity with two errors.



Parity bit

Data



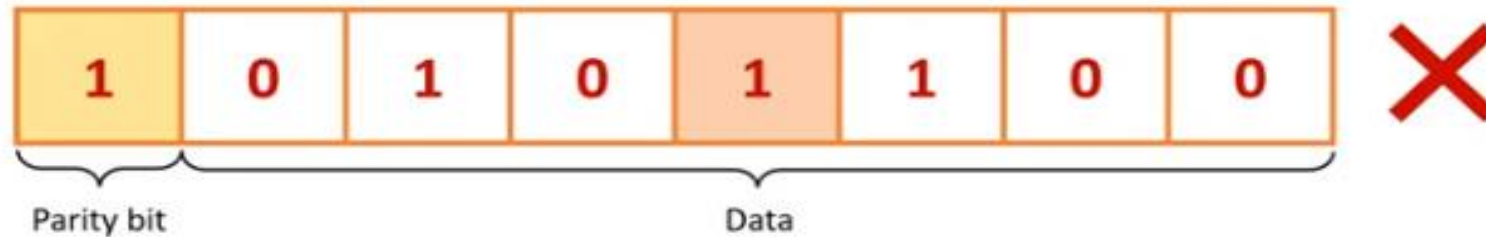
2.2 Methods of error detection

Parity block checks

We have seen how parity can be used to detect an error in a single byte of data.

However, the position of the error cannot be worked out.

A byte of data being sent using odd parity with a single error



Our only option is to ask for the data to be sent again.

However, if we send multiple bytes as a transmission block, not only can we spot that an error has occurred, but it can also be identified – and corrected by performing both horizontal and vertical parity checks on the whole block of data.

2.2 Methods of error detection

Parity block checks

1. Using even parity, a message is sent as a block of 8-bit bytes.
2. A horizontal parity check is performed on each byte, which identifies an error in one of the bytes.
3. Additional vertical parity checks are performed on the whole block.

Actual message received								
✓	0	1	0	0	0	0	0	1 A
✓	1	0	1	0	0	0	0	0
✓	0	1	0	1	0	1	1	0 V
✓	1	1	0	0	0	1	0	1 E
✓	1	1	0	1	0	0	1	0 R
✓	0	1	0	1	1	0	0	1 Y
✓	1	0	1	0	0	0	0	0
✓	1	1	0	1	0	1	0	0 T
✓	0	1	0	0	1	0	0	0 H
✓	1	1	0	0	1	0	0	1 I
✓	1	1	0	0	0	0	1	1 C
✓	0	1	0	0	1	0	1	1 K
✓	1	0	1	0	0	0	0	0
✗	1	1	0	0	0	1	0	0 D
✓	1	1	0	0	1	1	1	1 O
✓	0	1	0	0	0	1	1	1 G
✓	1	1	0	1	0	1	0	
✓	✓	✓	✓	✓	✓	✗	✓	

Parity block checks

1. Using even parity, a message is sent as a block of 8-bit bytes.
2. A horizontal parity check is performed on each byte, which identifies an error in one of the bytes.
3. Additional vertical parity checks are performed on the whole block.
4. The error can now be identified and corrected without having to ask for the data to be re-sent.

Actual message received								
✓	0	1	0	0	0	0	0	1 A
✓	1	0	1	0	0	0	0	0
✓	0	1	0	1	0	1	1	0 V
✓	1	1	0	0	0	1	0	1 E
✓	1	1	0	1	0	0	1	0 R
✓	0	1	0	1	1	0	0	1 Y
✓	1	0	1	0	0	0	0	0
✓	1	1	0	1	0	1	0	0 T
✓	0	1	0	0	1	0	0	0 H
✓	1	1	0	0	1	0	0	1 I
✓	1	1	0	0	0	0	1	1 C
✓	0	1	0	0	1	0	1	1 K
✓	1	0	1	0	0	0	0	0
✗	1	1	0	0	0	1	1	0 F
✓	1	1	0	0	1	1	1	1 O
✓	0	1	0	0	0	1	1	1 G
✓	1	1	0	1	0	1	0	
✓	✓	✓	✓	✓	✓	✗	✓	

2.2 Methods of error detection

2.2.1 Parity checking

- Before data is transferred, an agreement is made between sender and receiver regarding which of the two types of parity are used. This is an example of a **PROTOCOL**.
- **Activity 1**

Find the parity bits for each of the following bytes:

a 1 1 0 1 1 0 1

even parity used

b 0 0 0 1 1 1 1

even parity used

c 0 1 1 1 0 0 0

even parity used

d 1 1 1 0 1 0 0

odd parity used

e 1 0 1 1 0 1 1

odd parity used

2.2 Methods of error detection

2.2.1 Parity checking

- If a byte has been transmitted from 'A' to 'B', and even parity is used, an error would be flagged if the byte now had an odd number of 1-bits at the receiver's end.

- **Example 1 :**

sender's byte:

0	1	0	1	1	1	0	0
---	---	---	---	---	---	---	---

receiver's byte:

0	1	0	0	1	1	0	0
---	---	---	---	---	---	---	---

In this case, the receiver's byte has three 1-bits, which means it now has **odd parity** whilst the byte from the sender had **even parity** (four 1-bits). This clearly means an error has occurred during the transmission of the data.

2.2 Methods of error detection

2.2.1 Parity checking

- Before data is transferred, an agreement is made between sender and receiver regarding which of the two types of parity are used. This is an example of a **PROTOCOL**.
- **Activity 2**

Which of the following bytes have an error following data transmission?

a	1 1 1 0 1 1 0 1	even parity used
b	0 1 0 0 1 1 1 1	even parity used
c	0 0 1 1 1 0 0 0	even parity used
d	1 1 1 1 0 1 0 0	odd parity used
e	1 1 0 1 1 0 1 1	odd parity used

In each case where an error occurs, can you work out which bit is incorrect?

2.2 Methods of error detection

2.2.1 Parity checking : Example 1

- In this example, nine bytes of data have been transmitted. Agreement has been made that even parity will be used. Another byte, known as the **PARITY BYTE**, has also been sent. This byte consists entirely of the parity bits produced by the vertical parity check. The parity byte also indicates the end of the block of data. The following table shows how the data arrived at the receiving end:

	parity bit	bit 2	bit 3	bit 4	bit 5	bit 6	bit 7	bit 8
byte 1	1	1	1	1	0	1	1	0
byte 2	1	0	0	1	0	1	0	1
byte 3	0	1	1	1	1	1	1	0
byte 4	1	0	0	0	0	0	1	0
byte 5	0	1	1	0	1	0	0	1
byte 6	1	0	0	0	1	0	0	0
byte 7	1	0	1	0	1	1	1	1
byte 8	0	0	0	1	1	0	1	0
byte 9	0	0	0	1	0	0	1	0
parity byte	1	1	0	1	0	0	0	1

2.2 Methods of error detection

2.2.1 Parity checking

- A careful study of Table 1 shows the following:
 - byte 8 (row 8) has incorrect parity (there are three 1-bits)
 - bit 5 (column 5) also has incorrect parity (there are five 1-bits).
- First of all, the table shows that an error has occurred following data transmission.
- Secondly, at the intersection of row 8 and column 5, the position of the incorrect bit value (which caused the error) can be found.
- This means that byte 8 should have the value:

0	0	0	1	0	0	1	0
---	---	---	---	---	---	---	---

2.2 Methods of error detection

2.2.1 Parity checking

Which would also correct column 5 giving an even vertical parity (now has four 1-bits).

- This byte could therefore be corrected automatically as shown above, or an error message could be relayed back to the sender asking them to retransmit the block of data.
- If two of the bits change value following data transmission. it may be impossible to locate the error using the above method.
- **Parity checking: An even or odd number of bits set to 1 in a byte, used to check if the byte has been transmitted correctly**
- For example :

2.2 Methods of error detection

2.2.1 Parity checking

Activity 3

- 1 The following block of data was received after transmission from a remote computer; **odd parity** was being used by both sender and receiver. One of the bits has been changed during the transmission stage. Locate where this error is and suggest a corrected byte value:

	Parity bit	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Byte 1	0	1	1	0	0	0	1	0
Byte 2	1	0	1	1	1	1	1	1
Byte 3	1	0	0	1	1	0	0	0
Byte 4	0	1	1	0	1	0	1	0
Byte 5	1	1	1	0	0	1	1	0
Byte 6	1	0	0	0	0	1	0	1
Byte 7	0	1	1	1	0	0	0	0
Byte 8	0	0	0	0	0	0	0	1
Byte 9	0	1	1	1	1	0	1	0
Parity byte	1	0	1	1	1	1	0	0

2.2 Methods of error detection

2.2.1 Parity checking

Activity 4

- 2 The following block of data was received after transmission from a remote computer. Even parity was being used by both sender and receiver. One of the bytes has been changed during the transmission stage. Locate where this error is and suggest a corrected byte value.

	Parity bit	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Byte 1	1	1	0	0	0	0	0	0
Byte 2	0	0	1	1	1	1	0	0
Byte 3	0	1	0	0	0	1	1	1
Byte 4	1	0	1	0	1	1	1	1
Byte 5	0	0	0	1	0	0	0	1
Byte 6	0	0	1	1	1	1	1	1
Byte 7	1	0	1	1	0	1	0	0
Byte 8	0	1	0	1	0	0	0	1
Byte 9	1	1	1	0	0	1	0	0
Parity byte	0	0	1	1	1	0	0	1

2.2 Methods of error detection

2.2.2 Automatic Repeat Request (ARQ)

<https://www.youtube.com/watch?v=SqTS3ZlbwIA>

2.2 Methods of error detection

Automatic Repeat Query (ARQ)



2.2 Methods of error detection

2.2.2 Automatic Repeat Request (ARQ)

- **AUTOMATIC REPEAT REQUEST (ARQ)** also known as **Automatic Repeat Query** is another method used to check whether data has been correctly transmitted.
- It uses an **ACKNOWLEDGEMENT** (a message sent by the receiver indicating that data has been received correctly) and **TIMEOUT** (this is the time allowed to elapse before an acknowledgement is received).
- If an acknowledgement isn't sent back to the sender before timeout occurs, then the message is automatically resent.
- ARQ (Automatic Repeat Request) is an error control and packet recovery method for data transmission in which the receiver sends an alert to the sender if a packet is missing, so that the sender can resend the missing packet.

AUTOMATIC REPEAT REQUEST (ARQ): An error detection method that uses response and time out when transmitting data; if a response is not sent back to the sender in an agreed amount of time, then the data is re-sent.

2.2 Methods of error detection

2.2.2 Automatic Repeat Request (ARQ)



Automatic Repeat Query (ARQ)

Automatic Repeat Query (ARQ) is a method which can be used to verify that transmitted data has arrived error free, often used by mobile phone networks to ensure data integrity. It works as follows:

- The destination device receives the data transmission along with an error detection code.
- The code is used to determine if the data contains any transmission errors.
- If no errors have been detected, a positive acknowledgement is sent back.
- If an error is detected, a re-transmission request is sent back to the sending device.
- After a pre-determined amount of time, if the sending device hasn't received any form of acknowledge, it will automatically re-send the data.
- It will keep doing this until it either receives a positive acknowledgement or a pre-determined number of attempts to re-send the transmission have taken place.

2.2 Methods of error detection

2.2.3 Checksum

<https://www.youtube.com/watch?v=AtVWnyDDaDI>

Checksum



Carry

1	1	1	1	1			
1	0	0	0	0	1	0	0
0	0	1	0	0	1	0	0
1	1	1	0	0	0	1	0
1	0	0	1	1	0	0	1
0	0	1	0	0	0	1	1
						1	0
0	0	1	0	0	1	0	1
CHECKSUM	1	1	0	1	1	0	1

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Computer Networks

2.2 Methods of error detection

2.2.3 Checksum

Error checking and correction – Checksums

- **Checksums:** "A mathematical algorithm applied to a block of data."
- The data from the block is used in order to create the initial **checksum** which is then added and transmitted along with the original data.
- The same algorithm is applied at the other end, if the **checksums** match we consider the data to have been transmitted correctly.

0	1	0	1	1	1	1	1	1	1	0	0	0	0	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	1
95									140									40									181									
Checksum = 456																																				

2.2 Methods of error detection

2.2.3 Checksum

CHECKSUM: is another way to check if data has been changed or corrupted following data transmission. Data is sent in blocks and an additional value, the checksum, is also sent at the end of the block of data.

- To explain how this works, we will assume the checksum of a block of data is 1 byte in length. This gives a maximum value of $(2^8 - 1)$ (i.e. 255). **The value 0000 0000 is ignored in this calculation.**

Example 1 explains how a checksum is generated.

- If the sum of all the bytes in the transmitted block of data is ≤ 255 , then the checksum is this value.
- However, if the sum of all the bytes in the data block > 255 , then the checksum is found using the simple algorithm in Figure below .

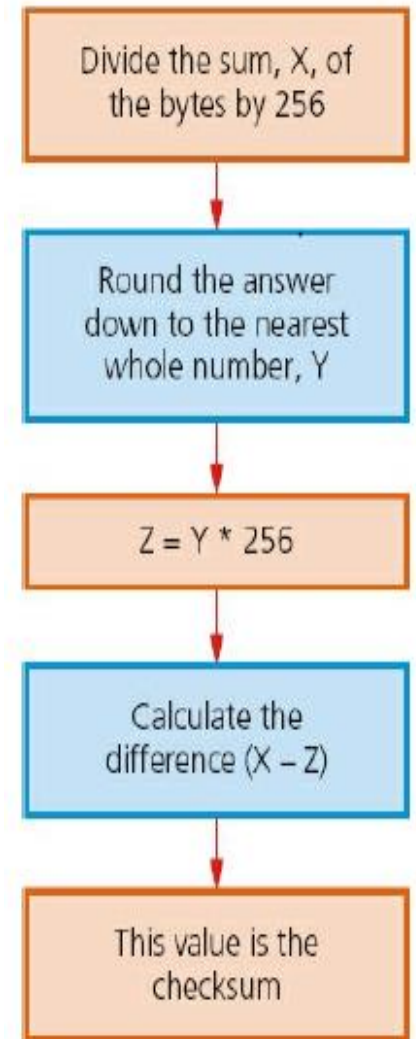
2.2 Methods of error detection

2.2.3 Checksum

- Suppose the value of X is **1185**, then tracing through the algorithm, we get:

$$X = 1185$$

- 1)- $1185/256 = 4.629$
- 2)- Rounding down to nearest whole number gives $Y = 4$
- 3)- Multiplying by 256 gives $Z = Y * 256 = 1024$
- 4)- The difference $(X - Z)$ gives the checksum: $(1185 - 1024) = 161$
- 5)- This gives the checksum = **161**



2.2 Methods of error detection

2.2.3 Checksum

When a block of data is about to be transmitted, the checksum for the bytes is first of all calculated. This value is then transmitted with the block of data. At the receiving end, the checksum is recalculated from the block of data received. This calculated value is then compared to the checksum transmitted. If they are the same value, then the data was transmitted without any errors; if the values are different, then a request is sent for the data to be retransmitted.

Checksum : A value transmitted at the end of a block of data; it is calculated using the other elements in the data stream and is used to check for transmission errors)

Activity 2.5

Activity 2.5

Calculate the checksum for blocks of data with the following byte sums:

- a 148
- b 905
- c 1450
- d 4095

2.2 Methods of error detection

2.2.4 Echo check

ECHO CHECK: when data is sent to another device, this data is sent back again to the sender. The sender compares the two sets of data to check if any errors occurred during the transmission process.

As you will have no doubt worked out, this isn't very reliable. If the two sets of data are different, it isn't known whether the error occurred when sending the data in the first place, or if the error occurred when sending the data back for checking!

However, if no errors occurred then it is another way to check that the data was transmitted correctly.

2.2 Methods of error detection



Error detection methods

Parity checks

- When sending a byte of data, we can use the Most Significant Bit as a parity bit.
- With even parity, we set the MSB to be 1 or 0 so that the total number of bits – including the parity bit – is even.
- With odd parity, we set the MSB to be 1 or 0 so that the total number of bits – including the parity bit – is odd.
- With this method, simple errors can be spotted, but there is no way to tell exactly where the error occurred.

Parity block checks

- When sending a block of data, parity checks can be performed both horizontally and vertically on the whole block.
- These checks can spot that an error has occurred *and* where, allowing the receiving device to fix the error.

Checksums

- A checksum is calculated by using an algorithm on the data to be transmitted.
- This checksum is added to the transmission and sent along with it.
- The receiving device re-calculates the checksum and compares the new one with the original.
- If the checksums do not match, it is assumed that an error has occurred

Echo check

- The receiving device returns an exact copy of the data it has just received.
- The data is compared to what was originally sent. If it is different, it is assumed that an error has occurred, and the original data is re-sent.

2.2 Methods of error detection

2.2.5 Check digits

https://www.youtube.com/watch?v=p60du0zr_H0

2.2 Methods of error detection

Check digits



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IGCSE
(0478/0984)

2.2 Methods of error detection

2.2.5 Check digits

Error checking and correction – Check digits

- **Check digits:** "A form of redundancy check used for error detection on identification numbers, such as bank account numbers, which are used in an application where they will at least sometimes be input manually."



The ISBN-10 number system used for books makes use of a process called 'Modulo-11' to create its **check digit**.

It becomes the last number.

2.2 Methods of error detection

2.2.5 Check digits

Error checking and correction – Check digits

- To check that a code number is valid, it is **NOT** necessary to recalculate the **check digit** completely.
- If the **check digit** itself is assigned a weight of 1, and the products of the digits (including the **check digit**) and their respective weights are calculated, their sum will be divisible by 11 if the check digit is correct.



2.2 Methods of error detection

2.2.5 Check digits

Method 1

Calculating a check digit

There are many check digit calculation methods. One such method, **Modulus 11**, is used by the ISBN-10 system for books. Let's calculate the check digit for the product code 817525765.

STEP 5: The remainder is then subtracted from 11 to give us the check digit, 2

Note that if the result of subtracting the remainder from 11 is 10, then the check digit is X.

So, the complete number with the check digit is: 8 1 7 5 2 5 7 6 5 **2**

Product code	8	1	7	5	2	5	7	6	5	
Weighting	10	9	8	7	6	5	4	3	2	
Multiply	80	9	56	35	12	25	28	18	10	
Addition	80	9	56	35	12	25	28	18	10	273
	Add all the numbers together									
Remainder	Remainder when the sum is divided by 11 e.g., 273 DIV 11 = 24 remainder 9									9
Subtraction	Subtract the remainder from 11 e.g., 11 – 9 = 2									2

2.2 Methods of error detection

2.2.5 Check digits

Method 1

Checking a check digit

The complete number with the check digit is: 8 1 7 5 2 5 7 6 5 **2**

STEP 3: The total is divided by 11. If the remainder is 0, the number is valid.

Product code	8	1	7	5	2	5	7	6	5	2	
Weighting	10	9	8	7	6	5	4	3	2	1	
Multiply	80	9	56	35	12	25	28	18	10	2	
Addition	80	9	56	35	12	25	28	18	10	2	275
	Add all the numbers together										
Remainder	Remainder when the sum is divided by 11 e.g., 275 DIV 11 = 25 remainder 0										0

2.2 Methods of error detection

2.2.5 Check digits

Method 2

Error checking and correction – Check digits

MODULO-11

STEP 1: Each digit of the code number is assigned a 'weight'. The right hand (least significant digit is given a weight of 2, the next digit to the left 3 and so on.

STEP 2: Each digit is multiplied by its weight and the products added together.

STEP 3: The sum of the products is divided by 11 and the remainder obtained.

STEP 4: The remainder is subtracted from 11.

STEP 5: The result is divided again by 11 and the remainder is the check digit.

Original code number

8 1 7 5 2 5 7 6 6

Weights

10 9 8 7 6 5 4 3 2

Multiply digit by its weight

80 9 56 35 12 25 28 18 12

Add products together

80 + 9 + 56 + 35 + 12 + 25 + 28 + 18 + 12

= 275

Divide by 11

11 remainder 0

Subtract remainder from 11

11 - 0 = 0

Divide by 11 again

0 remainder 0

Check digit = 0

ISBN 817525766-0 Check digit



2.2 Methods of error detection

2.2.5 Check digits



Check digits

A check digit is the last digit included in a code or other identification number.

- Calculated from all the other digits in the code.
- Commonly used with barcodes, ISBN numbers and Vehicle Identification Numbers (VIN).

Calculating a check digit:

- Starting from the left, give each digit a weighting.
- Multiply each digit by its weighting.
- Add the results together.
- Divide the total by 11.
- Subtract the remainder from 11 – the result is the check digit.

Checking a check digit:

- Starting from the left, give each digit a weighting – including the check digit.
- Multiply each digit by its weighting.
- Add the results together.
- Divide the total by 11.
- If the remainder is 0, the number is valid.

Key terms

- **Bit** – a 1 or a 0
- **Bit rate** – the rate at which data is transferred
- **Serial transmission** – data is sent along one wire, 1 bit at a time
- **Parallel transmission** – data is sent along multiple wires at the same time
- **Simplex transmission** – data can only be sent in one direction
- **Full duplex transmission** – data can be sent in both directions, at the same time
- **Half-duplex transmission** – data can be sent in both directions, but not at the same time
- **Checksum** – the number of bits being transmitted is sent with the data
- **Parity** – a 1 or a 0 is added to a binary number to make the number of bits odd (or even depending on the parity being used)
- **Check digit** – a calculation that is applied to data being transmitted; the calculation is performed before and after transmission and compared

IGCSE Past Paper Exam: Question 1:

1 The following questions all use even parity. Identify the parity bit required and add it to the end of each question.

a 101011

b 010000

c 0101001

d 11111111

e 01100101

f 010101010100

2 The following questions all use odd parity. Identify the parity bit required and add it to the end of each question.

a 010101

b 000011

c 111100

d 11000011

e 001111100

f 010101010

Parity checking: IGCSE Exam: Question 2

- (d) When eight bytes of data have been collected, they are transmitted to a computer 100 km away. Parity checks are carried out to identify if the data has been transmitted correctly. The system uses **even parity** and column 1 is the parity bit.

The eight bytes of data are sent together with a ninth parity byte:

	parity bit	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	1	0	0	0	0	1	0	0
byte 2	1	1	1	1	0	0	1	1
byte 3	0	1	0	0	1	0	0	0
byte 4	0	1	1	1	0	0	0	1
byte 5	1	0	0	0	1	1	1	1
byte 6	0	0	0	0	0	0	0	0
byte 7	1	1	1	0	1	0	0	0
byte 8	1	0	0	0	1	1	1	0
parity byte	1	0	1	1	0	1	1	1

- (i) Identify which of the eight bytes contains an error.

byte[1]

IGCSE Exam: Question 2

(ii) Identify which column contains an error.

column[1]

(iii) The incorrect bit is indicated where the byte number and column cross.

Give the corrected byte.

--	--	--	--	--	--	--	--

[1]

(iv) Calculate the denary value of the corrected byte.

.....

.....[1]

IGCSE Exam: Answer 2

- (d) When eight bytes of data have been collected, they are transmitted to a computer 100 km away. Parity checks are carried out to identify if the data has been transmitted correctly. The system uses **even parity** and column 1 is the parity bit.

The eight bytes of data are sent together with a ninth parity byte:

	parity bit	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	1	0	0	0	0	1	0	0
byte 2	1	1	1	1	0	0	1	1
byte 3	0	1	0	0	1	0	0	0
byte 4	0	1	1	1	0	0	0	1
byte 5	1	0	0	0	1	1	1	1
byte 6	0	0	0	0	0	0	0	0
byte 7	1	1	1	0	1	0	0	0
byte 8	1	0	0	0	1	1	1	0
parity byte	1	0	1	1	0	1	1	1

- (i) Identify which of the eight bytes contains an error.

byte (byte) 5 [1]

IGCSE Exam: Answer 2

(ii) Identify which column contains an error.

column (column) 4 [1]

(iii) The incorrect bit is indicated where the byte number and column cross.

Give the corrected byte.

[1 0 0 1 1 1 1 1] [1]

(iv) Calculate the denary value of the corrected byte.

..... 1 5 9
..... [1]

IGCSE Exam: Question 3

(a) Parity checks are often used to detect errors that may occur during data transmission.

The received bytes in the table below were transmitted using **odd parity**.

Tick (✓) to show whether each byte has been **corrupted during transmission** or **not corrupted during transmission**.

Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)
10110100		
01101101		
10000001		

[3]

IGCSE Exam: Question 4

- 8 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.
- (a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01100100	
10010001	
00000011	
10110010	

[4]

- (b) An error may **not** be detected when using a parity check.

Identify why an error may **not** be detected.

.....
..... [1]

IGCSE Exam: Question 4

(c) The data is sent using parallel half-duplex data transmission.

(i) Describe how data is sent using parallel half-duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) State **two** drawbacks of using parallel data transmission.

Drawback 1

.....

Drawback 2

..... [2]

IGCSE Exam: Answer 4

Question	Answer	Marks
8(a)	– Odd – Odd – Even – Even	4
8(b)	Any one from: – there is a transposition of bits – it does not check the order of the bits (just the sum of 1s/0s) – even number of bits change – incorrect bits still add up to correct parity	1
8(c)(i)	Four from: – Multiple bits are sent at the same time – Uses multiple wires – Data is sent in both directions ... – ... but only one direction at a time	4
8(c)(ii)	Any two from: – Bits may arrive skewed – More expensive to setup/manufacture/purchase cable – Limited distance – More prone to interference/error	2

IGCSE Exam: Question 5

- 6 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

- (a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01111100	
10010000	
10011001	
00101001	

[4]

- (b) The 8-bit binary value 10110001 is transmitted and received as 10110010

A parity check does **not** identify any errors in the binary value received.

State why the parity check does **not** identify any errors.

.....
..... [1]

IGCSE Exam: Question 5

(c) The data is sent using serial duplex data transmission.

(i) Describe how data is sent using serial duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) State **one** drawback of using serial data transmission, rather than parallel data transmission.

.....

..... [1]

IGCSE Exam: Answer 5

Question	Answer	Marks
6(a)	– Odd – Even – Even – Odd	4
6(b)	Any one from: – There is a transposition of bits – Bits still add up to correct parity	1
6(c)(i)	– Data is sent one bit at a time – Data is sent using a single wire – Data is sent in both direction ... – ... at the same time	4
6(c)(ii)	Any one from: – Data transmission can be slower (than parallel) – Additional data may need to be sent	1

IGCSE Exam: Question 6

- 9 Four 7-bit binary values are being transmitted from one computer to another. An odd parity check is being used to check for errors in the binary values.

Write the correct **Parity bit** for each **7-bit binary value** to make sure it meets **odd** parity.

Parity bit	7-bit binary value
.....	0000011
.....	1000000
.....	0111111
.....	1010101

[4]

IGCSE Exam: Answer 6

Question	Answer	Marks
9	- 1 - 0 - 1 - 1	4

IGCSE Exam: Question 7

- 9 The contents of three binary registers have been transmitted from one computer to another. **Even parity** has been used as an error detection method.

The outcome after transmission is:

Register A and **Register C** have been transmitted **correctly**.

Register B has been transmitted **incorrectly**.

Complete the **Parity bit** for each register to show the given outcome.

	Parity bit						
Register A		0	1	0	0	1	0
Register B		1	0	0	0	0	0
Register C		1	0	0	0	0	1

[3]

IGCSE Exam: Answer 7

PUBLISHED								
Question	Answer							Marks
9	1 mark per each correct parity bit:							3
Register A	Parity bit							
	1	0	1	0	0	1	0	1
Register B	1	1	0	0	0	0	0	1
Register C	1	1	0	0	0	0	1	1

IGCSE Exam: Question 8

- 5 The three binary numbers in the registers given have been transmitted from one computer to another.

One binary number has been transmitted incorrectly. This can be identified by the use of a **Parity bit**.

Identify the binary number that has been transmitted **incorrectly**. Explain how you identified the incorrect binary number.

	Parity bit							
Register A	1	0	1	1	1	0	0	1
Register B	1	1	1	0	0	1	1	1
Register C	1	0	0	1	1	0	1	1

IGCSE Exam: Question 8

The binary number that has been transmitted incorrectly is in **Register**

Explanation

.....

.....

.....

.....

..... [4]

IGCSE Exam: Answer 8

PUBLISHED

Question	Answer	Marks
5	- B Three from: - Added up the number of 1's / 0's in each register - With the parity bit, two registers have an odd number of 1's / 0's - One register has an even number of 1's / 0's - Odd parity must be the parity used	4

Automatic Repeat Request (ARQ): IGCSE Exam: Question 9

(b) Another method of error detection is Automatic Repeat reQuest (ARQ).

Explain how ARQ is used in error detection.

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..... [4]

Automatic Repeat Request (ARQ): IGCSE Exam: Question 10

- 2 Parity checks and Automatic Repeat reQuests (ARQ) can be used to check for errors during data transmission and storage.

(a) A system uses **even parity**. Write the appropriate parity bit for each byte.

Parity Bit							
	1	0	1	0	0	1	1
	1	0	1	1	1	1	1
	1	0	1	0	0	0	1

[2]

(b) Explain how Automatic Repeat reQuests (ARQ) are used in data transmission and storage.

.....
.....
.....
.....[2]

(c) State **one** other method that could be used to check for transmission errors.

.....[1]

Automatic Repeat Request (ARQ): IGCSE Exam: Answer 10

Question	Answer	Marks																																
2(a)	2 marks for 3 correct bits, 1 mark for 2 correct bits <table><tr><td>Parity Bit</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table>	Parity Bit								0	1	0	1	0	0	1	1	0	1	0	1	1	1	1	1	1	1	0	1	0	0	0	1	2
Parity Bit																																		
0	1	0	1	0	0	1	1																											
0	1	0	1	1	1	1	1																											
1	1	0	1	0	0	0	1																											
2(b)	Two from: ☐ Set of rules for controlling error checking/detection // it's an error detection method // used to detect errors ☐ Uses acknowledgement and timeout ☐ Request is sent (with data) requiring acknowledgement ☐ If no response/acknowledgment within certain time frame data package is resent ☐ When data received contains an error a request is sent (automatically) to resend the data ☐ The resend request is repeatedly sent until packet is received error free/limit is reached/acknowledgement received	2																																
2(c)	Checksum	1																																

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Question 11

- 7 Umar is a manager in an office. The data that he uses on a daily basis is stored on a file server close to his computer.

Data is sent from his computer to the file server using parallel simplex data transmission.

- (a) Describe how the data is sent using parallel simplex data transmission.

.....

.....

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.....

.....

..... [3]

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Question 11

- (b) The file server is moved to another building that is 1 km away.

Explain why the parallel simplex data transmission method that Umar uses is no longer suitable.

.....

.....

.....

..... [2]

- (c) Checksum and Automatic Repeat reQuest (ARQ) are both used when transmitting the data from a computer to the file server.

Explain why checksum and ARQ are both used.

.....

.....

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.....

..... [3]

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Answer 11

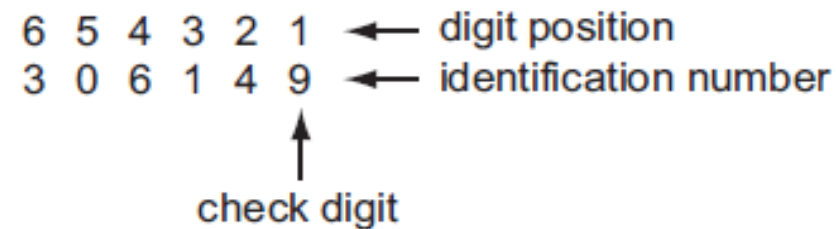
PUBLISHED

Question	Answer	Marks
7(a)	• Data is sent multiple bits at a time • ... down multiple wires • Data is sent in one direction only	3
7(b)	Any two from: • Parallel is not suitable for long distances // only suitable for short distances • ... the data could become skewed • Parallel cables are not manufactured above approx. 5m	2
7(c)	Any three from (MAX 2 for ARQ): • Checksum used to detect errors (during transmission) • ... using a calculated value • ARQ checks if data is received • ... uses acknowledgement and timeout • ... requests data be sent again if (checksum) detects error / not received	3

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

- 1 A company selling CDs uses a unique 6-digit identification number for each CD title. The right-most digit (position 1) is a *check digit*.

For example,



The validity of the number and check digit is calculated as follows:

- ☐ multiply **each** digit by its digit position
- ☐ add up the results of the multiplications
- ☐ divide the answer by 11
- ☐ if the remainder is 0, the identification number and check digit are valid.

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

- (a) Show whether the following identification numbers are valid or not. You **must** show how you arrived at your answer.

Identification number 1: 4 2 1 9 2 3

working:

.....

.....

.....

valid or not valid?

Identification number 2: 8 2 0 1 5 6

working:

.....

.....

.....

valid or not valid? [3]

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

(b) Find the check digit for this identification number.

5 0 2 4 1 ____

working:

.....

.....

.....

check digit: [2]

(c) Describe, with examples, **two** different types of data entry errors that a check digit would detect.

1

.....

2

..... [2]

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Answer 12

- 1 (a) 1 mark for the correct working in BOTH parts
1 mark for valid
1 mark for not valid

Identification number 1: working

$$\begin{aligned} &= (4 \times 6) + (2 \times 5) + (1 \times 4) + (9 \times 3) + (2 \times 2) + (3 \times 1) \\ &= 24 + 10 + 4 + 27 + 4 + 3 \\ &= 72 \div 11 \\ &= 6 \text{ remainder } 6 \\ &\text{valid/not valid: NOT valid} \end{aligned}$$

Identification number 2: working

$$\begin{aligned} &= (8 \times 6) + (2 \times 5) + (0 \times 4) + (1 \times 3) + (5 \times 2) + (6 \times 1) \\ &= 48 + 10 + 0 + 3 + 10 + 6 \\ &= 77 \div 11 \\ &= 7 \text{ remainder } 0 \\ &\text{valid/not valid: VALID} \end{aligned}$$

[3]

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Answer 12

(b) 1 mark for correct working + 1 mark for check digit

working

$$\begin{aligned} &= (5 \times 6) + (0 \times 5) + (2 \times 4) + (4 \times 3) + (1 \times 2) \\ &= 30 + 0 + 8 + 12 + 2 \\ &= 52 \end{aligned}$$

need to add 3 to make the total 55 (i.e. exactly divisible by 11)

check digit: 3

[2]

(c) 1 mark for each description and example

2 digits transposed

(e.g. 280419 becomes 280149/two digits have been switched)

incorrect digit

(e.g. 280419 becomes 250419/one of the digits has been mistyped)

[2]

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 13

There are various methods used to detect errors that can occur during data transmission and storage.

Describe each of the following error detection methods.

Parity check

.....
.....
.....

Check digit

.....
.....
.....

Checksum

.....
.....
.....

Automatic Repeat request (ARQ)

.....
.....
.....

[8]

To know more about check digit please click on Link : <https://www.bbc.co.uk/bitesize/guides/zfspfcw/revision/6>

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 14

Identify and describe two methods of error checking that can be used to make sure that the data stored after transmission is accurate.

Method 1

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.....

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.....

.....

Method 2

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[6]

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 15

Five computer terms and seven descriptions are shown below.

Draw a line to connect each computer term to its correct description.

Computer term

Serial, simplex data transmission

Parallel, half-duplex data transmission

Parity check

Automatic repeat request (ARQ)

Checksum

Description

Several bits of data sent down several wires, in both directions, but not at the same time

Several bits of data sent down several wires, in both directions, at the same time

An even or odd number of bits set to 1 in a byte, used to check if the byte has been transmitted correctly

One bit sent at a time, over a single wire in one direction only

An additional digit placed at the end of a number to check if the number has been entered correctly

A value transmitted at the end of a block of data; it is calculated using the other elements in the data stream and is used to check for transmission errors

An error detection method that uses response and time out when transmitting data; if a response is not sent back to the sender in an agreed amount of time, then the data is re-sent

IGCSE Exam: Question 16

9 (a) Computers can transmit data using different methods.

Describe the **three** data transmission methods given.

(i) Serial data transmission

.....

.....

.....

.....[2]

(ii) Parallel data transmission

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.....

.....

.....[2]

(iii) Duplex data transmission

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.....

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.....[2]

IGCSE Exam: Question 16

- (b) Data can sometimes be corrupted when it is transmitted from one computer to another, causing errors to be present in the data.

Identify and describe **three** methods of error detection that could be used to see if an error has occurred.

Error detection method 1

Description

.....

.....

.....

Error detection method 2

Description

.....

.....

.....

Error detection method 3

Description

.....

.....

.....

[9]

IGCSE Exam: Answer 16

Question	Answer	Marks
9(a)(i)	Two from: ☐ Data is transmitted one bit at a time ☐ Data is transmitted using a single wire ☐ Bits arrive in order/sequence	2
9(a)(ii)	Two from: ☐ Data is transmitted multiple bits at a time/simultaneously ☐ Data is transmitted using multiple wires ☐ Bits may arrive out of sequence/skewed (and are reordered)	2
9(a)(iii)	1 mark for each: ☐ Data is transmitted in both directions ☐ ... at the same time/simultaneously	2

IGCSE Exam: Answer 16

Question	Answer	Marks
9(b)	Maximum of three marks per error detection method. 1 mark for naming the method, 2 marks for describing it. Parity (check) ☐ Odd or even parity can be used ☐ Bits are added together // 1 bits are counted ☐ Parity bit added (depending on parity set) ☐ Parity checked on receipt ☐ If parity bit is incorrect an error is detected Checksum ☐ Calculation performed on data (to get the checksum) ☐ Checksum sent with data ☐ Checksum recalculated after transmission ☐ Comparison made between checksum before and checksum after transmission ☐ Error detected if checksums are different Automatic repeat request (ARQ) ☐ Uses acknowledgement and timeout ☐ Request is sent (with data) requiring acknowledgement ☐ If no response/acknowledgment within certain time frame data package is resent ☐ When data received contains an error a request is sent (automatically) to resend the data ☐ The resend request is repeatedly sent until packet is received error free/limit is reached/acknowledgement received	9

IGCSE Exam: Question 17

- 5 The three binary numbers in the registers X, Y and Z have been transmitted from one computer to another.

								Parity bit
Register X	1	0	0	1	0	0	1	0
Register Y	1	1	1	0	0	1	1	1
Register Z	1	1	1	0	1	0	0	1

Only **one** binary number has been transmitted correctly. This is identified through the use of a parity bit.

Identify which register contains the binary number that has been transmitted **correctly**. Explain the reason for your choice.

The binary number that has been transmitted correctly is in **Register**

Explanation

.....

.....

.....

IGCSE Exam: Answer 17

PUBLISHED

Question	Answer	Marks
5	1 mark for correct register, 3 marks for reason: - Register Y Any three from: - Count the number of 1/0 bits (in each byte/register) - Two bytes/registers have an odd number of 1/0 bits // Two have odd parity - Even parity must be the parity used - One byte/register has an even number of 1/0 bits // One uses even parity - The two with an odd number of one bits/odd parity are incorrect // Register X and Z should have even parity	4

IGCSE Exam: Question 18

4 A company transmits data to external storage at the end of each day.

(a) Parity checks can be used to check for errors during data transmission.

The system uses **odd parity**.

(i) Tick (✓) to show for each of the received bytes whether they have been **transmitted correctly** or **transmitted incorrectly**.

Received byte	Transmitted correctly (✓)	Transmitted incorrectly (✓)
10001011		
10101110		
01011101		
00100101		

[4]

(ii) State **one** other method that could be used to check for transmission errors.

..... [1]

IGCSE Exam: Question 18

(b) Data can be transferred using parallel or serial data transmission.

(i) Describe what is meant by parallel data transmission.

.....

.....

.....

..... [2]

(ii) Give **one** application of parallel data transmission.

.....

..... [1]

IGCSE Exam: Answer 18

Question	Answer			Marks
4(a)(i)	Received Byte	Transmitted correctly (✓)	Transmitted incorrectly (✓)	4
	10001011		✓	
	10101110	✓		
	01011101	✓		
	00100101	✓		
4(a)(ii)	One from: - ARQ - Check Sum			1

IGCSE Exam: Question 19

- 5 (a) Parity checks are often used to detect errors that may occur during data transmission.

The received bytes in the table below were transmitted using **odd parity**.

Tick (✓) to show whether each byte has been **corrupted during transmission** or **not corrupted during transmission**.

Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)
10110100		
01101101		
10000001		

[3]

IGCSE Exam: Question 19

(b) Another method of error detection is Automatic Repeat reQuest (ARQ).

Explain how ARQ is used in error detection.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

IGCSE Exam: Answer 19

Question	Answer	Marks												
5(a)	1 mark per correct tick <table> <tr> <th>Received byte</th><th>corrupted during transmission (✓)</th><th>not corrupted during transmission (✓)</th></tr> <tr> <td>10110100</td><td>✓</td><td></td></tr> <tr> <td>01101101</td><td></td><td>✓</td></tr> <tr> <td>10000001</td><td>✓</td><td></td></tr> </table>	Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)	10110100	✓		01101101		✓	10000001	✓		3
Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)												
10110100	✓													
01101101		✓												
10000001	✓													
5(b)	Four from: ☐ Uses acknowledgement and time out ☐ Check performed on received data // error is detected by e.g. parity check, check sum ☐ If error detected, request sent to resend data // negative acknowledgment is used ☐ If no acknowledgement is sent that data is received // positive acknowledgment is used ☐ Data is resent / Resend request repeated, till data is resent correctly ... ☐ ... or request times out // limit is reached	4												

IGCSE Exam: Question 20

9 Check digits are used to ensure the accuracy of entered data.

A 7-digit number has an extra digit on the right, called the check digit.

digit position:	1	2	3	4	5	6	7	8
digit:	—	—	—	—	—	—	—	—
								↑ check digit

The check digit is calculated as follows:

- each digit in the number is multiplied by its digit position
- the seven results are then added together
- this total is divided by 11
- the remainder gives the check digit (if the remainder = 10, the check digit is X)

(a) Calculate the check digit for the following number. Show all your working.

4 2 4 1 5 0 8 ...

.....

.....

.....

Check digit

[2]

IGCSE Exam: Question 20

(b) An operator has just keyed in the following number:

3 2 4 0 0 4 5 X

Circle below **correct** if the check digit is correct **OR incorrect** if the check digit is incorrect.

correct

incorrect

Explain your answer.

.....

.....

.....

.....

[3]

IGCSE Exam: Answer 20

9 (a) 1 mark for correct check digit and 1 mark for showing the calculation

$$(4 \times 1) + (2 \times 2) + (4 \times 3) + (1 \times 4) + (5 \times 5) + (0 \times 6) + (8 \times 7)$$

$$= 4 + 4 + 12 + 4 + 25 + 0 + 56 = 105$$

$$105/11 = 9 \text{ remainder } 6$$

check digit is: 6

1 mark for any correct
line of working

[2]

(b) incorrect check digit

[1]

– check digit should be 1

– $(3 \times 1) + (2 \times 2) + (4 \times 3) + (0 \times 4) + (0 \times 5) + (4 \times 6) + (5 \times 7) // 3 + 4 + 12 + 0 + 0 + 24 + 35 //$
Total = 78

– 78/11 gives 7 remainder 1

[2]

IGCSE Exam: Question 21

- 4 Nine bytes of data are transmitted from one computer to another. Even parity is used. An additional parity byte is also sent.

The ten bytes arrive at the destination computer as follows:

	parity bit	bit 2	bit 3	bit 4	bit 5	bit 6	bit 7	bit 8
byte 1	1	1	1	0	1	1	1	0
byte 2	0	0	0	0	0	1	0	1
byte 3	0	1	1	1	1	0	0	0
byte 4	1	1	0	0	0	0	0	0
byte 5	1	0	1	1	1	1	1	0
byte 6	0	1	0	1	1	0	0	1
byte 7	0	1	1	1	0	0	1	1
byte 8	0	0	1	1	0	1	1	0
byte 9	1	1	0	0	0	0	1	1
parity byte	0	0	1	0	0	0	1	0

One of the bits was corrupted during the data transmission.

- (a) **Circle** the corrupt bit in the corrupt byte in the table above.

[1]

IGCSE Exam: Question 21

(b) Explain how the corrupted bit was found.

.....

.....

.....

.....

.....

.....[2]

IGCSE Exam: Answer 21

4 (a) Intersection of Row 7 and column 4 circled [1]

- (b) – Row (byte number) 7 has an odd number of 1s (five 1s)
– Column (bit number) 4 has an odd number of 1s (five 1s)

[2]

IGCSE Exam: Question 22

9 In the following barcode, each binary number is made up of seven bars.

Each bar is black or grey.

A black bar is interpreted as a “1” and a grey bar is interpreted as a “0”.

(a) Write the binary numbers that would be produced from this barcode:



Binary number A Binary number B

Binary number A:

--	--	--	--	--	--	--

Binary number B:

--	--	--	--	--	--	--

[2]

IGCSE Exam: Question 22

(b) This barcode system uses odd parity.

Write the parity bit for each of the binary numbers in **part (a)**:

	Parity bit
Binary number A:	☐
Binary number B:	☐

[2]

IGCSE Exam: Answer 22

9 (a)

Binary number A:

1	1	1	0	0	1	0
---	---	---	---	---	---	---

Binary number B:

1	0	0	1	1	1	0
---	---	---	---	---	---	---

[2]

(b)

Parity Bit

Binary number A

1

Binary number B

1

[2]

IGCSE Exam: Question 23

5 Parity checks are often used to check for errors that may occur during data transmission.

(a) A system uses **even parity**.

Tick (✓) to show whether the following three bytes have been transmitted correctly or incorrectly.

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		
0 1 1 1 1 1 0 0		
0 1 1 0 1 0 0 1		

[3]

IGCSE Exam: Question 23

(b) A parity byte is used to identify which bit has been transmitted incorrectly in a block of data.

The word "F L O W C H A R T" was transmitted using nine bytes of data (one byte per character). A tenth byte, the parity byte, was also transmitted.

The following block of data shows all ten bytes received after transmission. The system uses **even parity** and column 1 is the parity bit.

	letter	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	F	1	0	1	0	0	1	1	0
byte 2	L	1	0	1	0	1	1	0	0
byte 3	O	1	0	1	0	1	1	1	1
byte 4	W	1	0	1	1	0	1	1	1
byte 5	C	1	0	1	0	0	0	1	1
byte 6	H	0	0	1	0	1	0	0	0
byte 7	A	0	0	1	0	0	1	0	1
byte 8	R	1	0	1	1	0	0	1	0
byte 9	T	1	0	1	1	0	1	0	0
parity byte		1	0	1	1	1	1	1	0

(i) One of the bits has been transmitted incorrectly.

Write the byte number and column number of this bit:

Byte number

Column number

[2]

IGCSE Exam: Question 23

(ii) Explain how you arrived at your answer for part (b)(i).

.....

.....

.....

.....[2]

(c) Give the denary (base 10) value of the byte: **1 0 1 1 1 1 1 0**

.....

.....[1]

(d) A parity check may not identify that a bit has been transmitted incorrectly.

Describe **one** situation in which this could occur.

.....

.....

.....[1]

IGCSE Exam: Answer 23

5 (a) 1 mark per correctly placed tick

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		✓
0 1 1 1 1 1 0 0		✓
0 1 1 0 1 0 0 1	✓	

[3]

(b) (i) byte number: 7

column number: 6

[2]

IGCSE Exam: Answer 23

(ii) Any **two** from:

- letter “A”(byte 7) transmitted as odd parity (three 1s)
- column 6 has odd parity (seven 1s)
- intersection of byte 7 and column 6 indicates incorrect bit value

[2]

(c) 190

[1]

(d) Any **one** from:

- 2 bits interchanged (e.g. $1 \rightarrow 0$ and $0 \rightarrow 1$) that won't change parity value
- even number of bits/digits are transposed
- If there are multiple errors in the same byte/column, that still produce the same parity bit, the error will not be detected

[1]

IGCSE Exam: Question 24

12 Parity checks are used to check for errors during data transmission. A system uses odd parity.

(a) Complete the following two bytes of data so that they both have odd parity:

	1	1	1	1	0	0	0
	0	0	0	0	1	1	1

[2]

(b) Name and describe another method which can be used to check whether data has been correctly transmitted.

Name of method

Description

.....

.....

[2]

IGCSE Exam: Answer 24

12 (a)

1	1	1	1	1	0	0	0
0	0	0	0	0	1	1	1

[2]

(b) 1 mark for error detection method and 1 mark for description

- Check sum
- ... sum of bits is transmitted and checked against the sum of the received bits
- Check digit
- ... a digit that is calculated (e.g. using modulo-11) and transmitted with the data
- ARQ
- ... when an error is detected in a packet of data a request is automatically sent for the data to be resent

[2]

IGCSE Exam: Question 25

- 7 (a) Check digits are used to ensure the accuracy of input data.

A 7-digit code number has an extra digit on the right, called the check digit.

Digit position	1	2	3	4	5	6	7	8
Digit	—	—	—	—	—	—	—	—

The check digit is calculated as follows:

- each digit in the number is multiplied by its digit position
- the seven results are then added together
- this total is divided by 11
- the remainder gives the check digit (if the remainder = 10, the check digit is X)

- (i) Calculate the check digit for the following code number. Show all your working.

4 2 4 1 5 0 8 ...

.....

.....

.....

Check digit

[2]

IGCSE Exam: Question 25

(ii) An operator has just keyed in the following code number:

3 2 4 0 0 4 5 X

Has the operator correctly keyed in the code number?

.....

Give a reason for your answer.

.....

.....

.....

.....

[3]

IGCSE Exam: Question 25

(b) When data are transmitted from one device to another, a parity check is often carried out on each byte of data. The parity bit is often the leftmost bit in the byte.

(i) If a system uses even parity, give the parity bit for each of the following bytes:

parity bit

	1	1	0	0	1	1	0
--	---	---	---	---	---	---	---

parity bit

	0	0	0	0	0	0	1
--	---	---	---	---	---	---	---

[2]

(ii) A parity check can often detect corruption of a byte.

Describe a situation in which it **cannot** detect corruption of a byte.

.....
.....
.....[1]

IGCSE Exam: Answer 25

7 (a) (i) 1 mark for correct check digit and 1 mark for showing the calculation

$$(4 \times 1) + (2 \times 2) + (4 \times 3) + (1 \times 4) + (5 \times 5) + (0 \times 6) + (8 \times 7)$$

$$= 4 + 4 + 12 + 4 + 25 + 0 + 56 = 105$$

$$105/11 = 9 \text{ remainder } 6$$

check digit is: 6

[2]

(ii) 1 mark

– No/incorrect check digit

2 marks

- Total is 78
- $78/11 \dots$
- $\dots$ gives 7 remainder 1
- check digit should be 1

[3]

IGCSE Exam: Answer 25

(b) (i) 1 mark for each correct parity bit

parity bit

0	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---

parity bit

1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

[2]

(ii) Any **one** from:

- an even number of digits are changed
- a transposition error(s) has occurred

[1]

Learning Objectives

2.3 Encryption

Candidates should be able to:

- 1 Understand the need for and purpose of encryption when transmitting data
- 2 Understand how data is encrypted using symmetric and asymmetric encryption

Notes and guidance

- Asymmetric encryption includes the use of public and private keys

2.3 Encryption

<https://www.youtube.com/watch?v=Ih7BT2uFKh4>

2.3 Encryption

Symmetric and
asymmetric encryption



CAMBRIDGE
IGCSE
(0478/0984)

2.3 Encryption

Understanding Encryption!

<https://www.youtube.com/watch?v=1y1M2fZqIIQ>



2.3 Encryption

What is encryption?

Encryption: *The process of encoding a message so it can be read only by the sender and the intended recipient.*

Encryption has been around for as long as we have been producing written communications.



2.3 Encryption

What is encryption?

Encryption is a process that scrambles readable text so it can only be read by the person who has the secret code, or decryption key. It helps provide data security for sensitive information.

Vast amounts of personal information are managed online and stored in [the cloud](#) or on servers with an ongoing connection to the web. It's nearly impossible to do business of any kind without your personal data ending up in an organization's networked computer system, which is why it's important to know how to help keep that data private.

How does encryption work?

Encryption takes plain text, like a text message or email, and scrambles it into an unreadable format — called “cipher text.” This helps protect the confidentiality of digital data either stored on computer systems or transmitted through a network like the Internet.

When the intended recipient accesses the message, the information is translated back to its original form. This is called decryption.

To unlock the message, both the sender and the recipient have to use a “secret” encryption key — a collection of algorithms that scramble and unscramble data back to a readable format

2.3 Encryption

Simple encryption using Caesar cipher

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Shift of 5 places
Key = 5

V W X Y Z A B C D E F G H I J K L M N O P Q R S T U

- One of the first forms of **encryption** was the Caesar cipher, which replaces each letter of the alphabet with another letter a fixed distance away from the original letter.
- The aim of **encryption** is to make the original message impossible to crack without the key.



C	O	M	P	U	T	I	N	G
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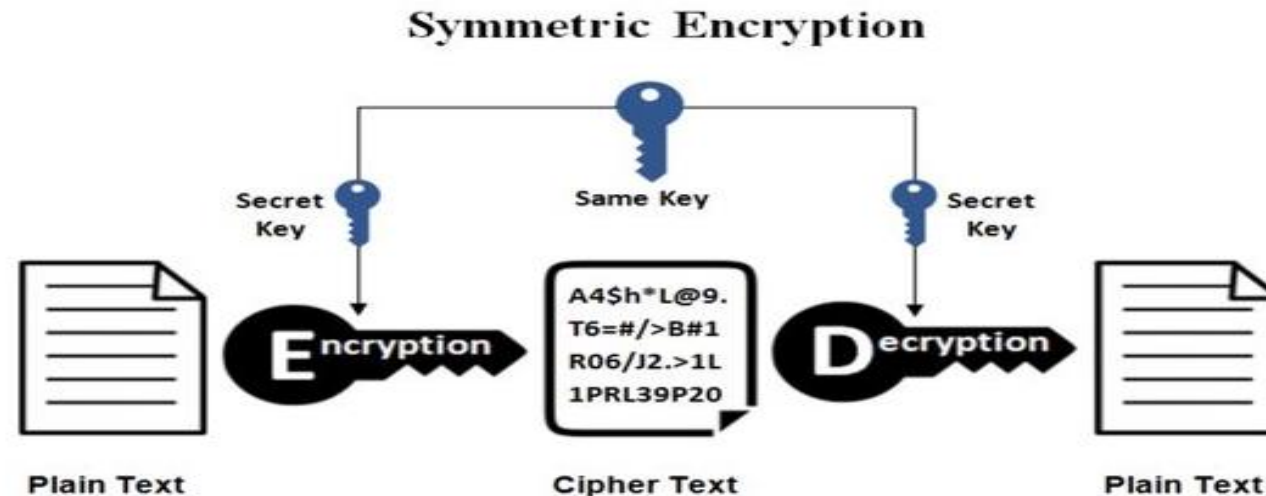
X	J	H	K	P	O	D	I	B
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2.3 Encryption

Symmetric Encryption

Symmetric encryption uses a single key to encrypt and decrypt. If you encrypt a zip file, then decrypt with the same key, you are using symmetric encryption. Symmetric encryption is also called “secret key” encryption because the key must be kept secret from third parties.

Again Symmetric cryptography, known also as secret key cryptography, is the use of a single shared secret to share encrypted data between parties. Ciphers in this category are called symmetric **because you use the same key to encrypt and to decrypt the data.**



2.3 Encryption

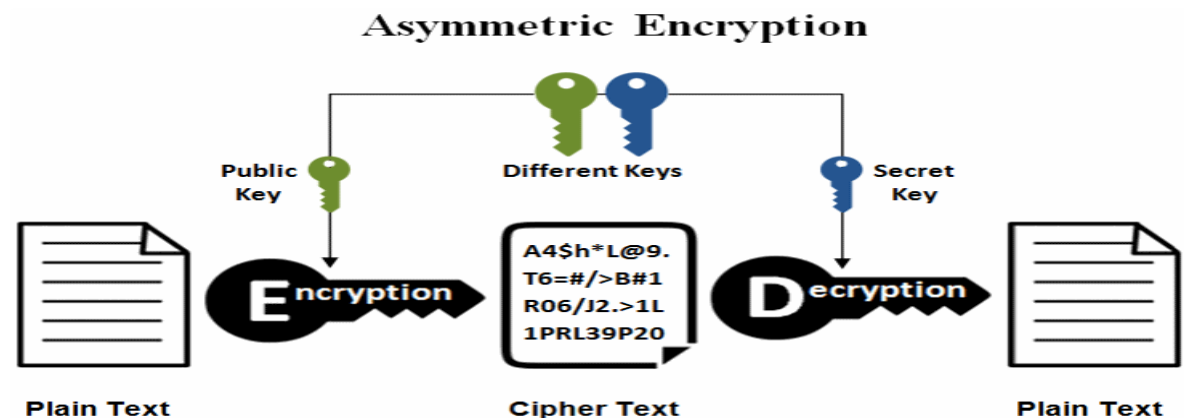
Asymmetric Encryption

Asymmetric cryptography, also known as public-key cryptography, is a **process that uses a pair of related keys one public key and one private key to encrypt and decrypt a message and protect it from unauthorized access or use.**

A [public key](#) is a cryptographic key that can be used by any person to encrypt a message so that it can only be decrypted by the intended recipient with their private key. A private key -- also known as a secret key -- is shared only with key's initiator.

When someone wants to send an encrypted message, they can pull the intended recipient's public key from a public [directory](#) and use it to encrypt the message before sending it. The recipient of the message can then decrypt the message using their related [private key](#).

If the sender encrypts the message using their private key, the message can be decrypted only using that sender's public key, thus authenticating the sender. These encryption and decryption processes happen automatically; users do not need to physically lock and unlock the message.



2.3 Encryption



Symmetric and asymmetric encryption

Encryption is the process of encoding a message so it can only be read by the sender and intended recipient – this is especially useful when transmissions including personal or sensitive information.

Symmetric encryption:

- Uses the same (pre-shared) private key to encode and decode data.
- This key must be transmitted (shared) in some way prior to sending an encrypted message.
- However, shared private keys can be intercepted, making this method less secure.

Asymmetric encryption:

- Uses two separate but related keys.
- The public key is shared with others and used by the recipient to encrypt a message.
- In this way, a message can only be deciphered by the intended recipient with their private key.
- Private keys must never be shared.

IGCSE Past Paper Exam – Question 1

- 3** A finance company is concerned that its employees are being distracted by using gaming websites at work.

(a) Explain how a firewall could help prevent this distraction.

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..... [4]

IGCSE Past Paper Exam – Question 1

- (b)** The finance company is also worried about the security of the data stored on its servers.

The company has decided to encrypt the data to improve the security.

Describe how the data are encrypted.

..... [4]

IGCSE Past Paper Exam – Question 1

- (c)** The finance company realises that its computer systems have been hacked.

The company thinks that spyware was used to obtain a user's password.

Explain how spyware could have been used to obtain the user's password.

[6]

IGCSE Past Paper Exam – Answer 1

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Question	Answer	Marks
3(a)	Four from: – The company could use the firewall to set criteria – Gaming websites can be listed as blocked websites // ports can be blocked – The firewall would examine any traffic leaving the network – If it detected traffic requesting a listed website, it will block access to it – Keeps a log of all attempts to access blocked websites	4
3(b)	Four from: – An encryption algorithm is used – ... to scramble data – The original data is called the plain text – A key is used to encrypt the data – The key is applied to the plain text – Plain text is encrypted into cypher text	4
3(c)	Six from: – The user could have been sent an email with an attachment / link containing the spyware – The user could have clicked a link on an untrusted website – When the attachment / link was clicked the spyware was downloaded onto the user's computer – The spyware recorded all the key logs from the user's keyboard – The recorded key logs were sent back to the creator of the spyware – The key logs were analysed – A common pattern / word in the key logs could have allowed a password to be identified	6

IGCSE Past Paper Exam – Question 2

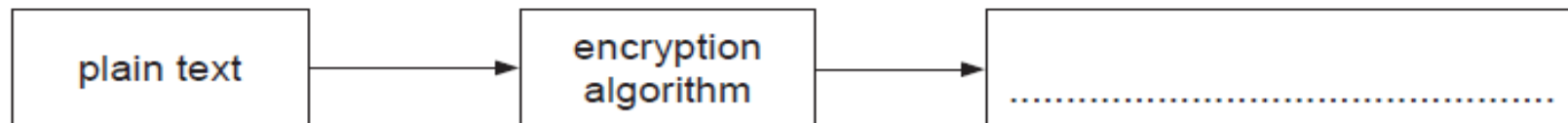
6 (a) State what is meant by encryption.

.....
..... [1]

(b) State what is meant by symmetric encryption.

.....
..... [1]

(c) Complete the diagram:



[1]

IGCSE Past Paper Exam – Answer 2

6 (a) Any **one** from:

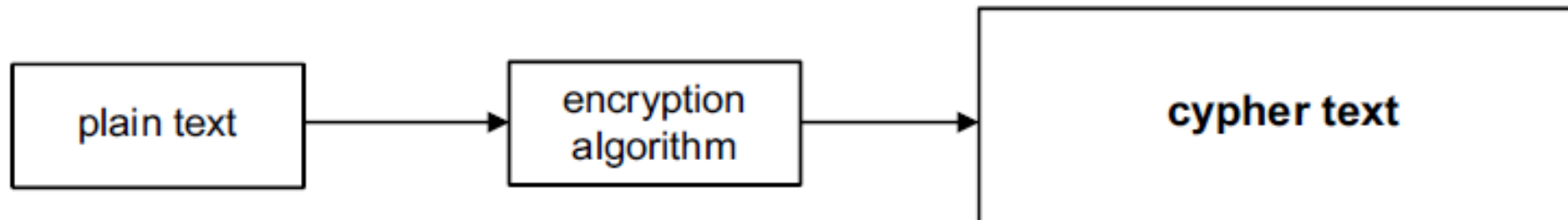
- jumbling up/scrambling characters so that message makes no sense
- requires an encryption key to encrypt data
- need decryption key to decipher encrypted message

[1]

(b) Uses the same key to encrypt and decrypt message

[1]

(c) 1 mark for correct name in box



[1]

IGCSE Past Paper Exam – Question 3

- 7 Adele chooses to set a biometric password for her mobile device, instead of a personal identification number (PIN).

(a) State what is meant by a biometric password.

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..... [1]

(b) Give **two** reasons why a biometric password is more secure than a PIN.

Reason 1
.....
Reason 2
..... [2]

IGCSE Past Paper Exam – Question 3

(c) Adele has a software-based firewall installed on her mobile device.

The firewall gathers data about the traffic coming into and going out of her mobile device.

Explain how the firewall uses the gathered data to keep the mobile device more secure.

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.....
..... [3]

(d) Adele also encrypts the data on her mobile device to keep it more secure.

State how encryption will keep the data more secure.

.....
..... [1]

IGCSE Past Paper Exam – Answer 3

Question	Answer	Marks
7(a)	Any one from: • Uses biological data • It uses characteristics/features that belong to a human	1
7(b)	Any two from: • A biometric password cannot be guessed • It is very difficult to fake a biometric password • A biometric password cannot be recorded by a keylogger/spyware • A perpetrator cannot shoulder surf to see a biometric password	2

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Question	Answer	Marks
7(c)	Any three from: • The traffic/data is compared to set criteria/rules/whitelist/blacklist • If the traffic/data does/does not meet the criteria/rules/whitelist/blacklist it will be rejected/blocked ... • ... and an alert can be sent to warn Adele • ... this can help prevent hackers/malicious software (gaining access to the data)	3
7(d)	• It makes the data meaningless	1

IGCSE Past Paper Exam – Question 4

(c) Data is encrypted using 128-bit symmetric encryption before it is transmitted.

(i) Explain what is meant by encryption.

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..... [2]

(ii) State how the strength of the encryption can be improved.

.....

..... [1]

IGCSE Past Paper Exam – Answer 4

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Question	Answer	Mark
2(c)(i)	Any two from: • Scrambles data • ... making it meaningless/unintelligible • Uses an algorithm / key • Data / plain text is changed to cipher text	2
2(c)(ii)	Any one from: • Increase the length of the key // use more than 128 bits • Uses a more complex encryption algorithm	1